# An Omitted Mode Is a Rare Rule: The Sampling-Verification Danger Law in Continuous Code World Models

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**Abstract**

In the Code World Model (CWM) paradigm, an LLM synthesizes an executable world model that a classical planner searches, accepted when it matches sampled transitions. In discrete games, a companion paper showed this gate certifies the wrong thing: a model can pass at 100% transition accuracy and still lose systematically, following danger = play_cost × $(1 - \text{rarity})^N$ (Aguilar Martín, 2026). This paper moves to continuous state spaces, where model-based RL treats world-model error as *pervasive*, not localized. The law transfers (its ingredients are measure-theoretic) and we prove smooth (Lipschitz) pairs cannot realize the *exactly* localized verified-but-wrong geometry (Section 3): the premise is realizable exactly where systems are hybrid, so an omitted mode is a rare rule. On two 1D hybrid instruments the gate misses the mode at the predicted $(1 - r)^N$ rate and the mode-blind planner is exploited below random at every knob once the phantom reward's tail is removed, play_cost knob-invariant to $10^{-4}$; a distrust-region planner collapses this to a bounded first-contact transient. With real LLM synthesis, every sample missing the mode yielded a certified, blind, exploited model; when the sample contained the mode, GPT-5.x repaired the exact rule (106/106) — unlike the discrete setting. But repair is geometry-dependent: on a 4D bi-modal instrument with 2D modes it inverts. Neither GPT-5.x size recovers the region rule in any of **156** mode-containing seeds (disc, zero-curvature square, 3×-budget guidance), and a Claude arm reproduces the same failure classes. The mechanism is a *template prior* over low-complexity regions (half-planes on the disc, discs on the square), not boundary curvature, so the law's exhaustiveness is geometry-scoped; certification stays sound. Identifiability is a sample property; repair is model- and geometry-dependent. Localization — both the danger and the repair — is a representational property of *code*.

## 1 Introduction

The companion paper (Aguilar Martín, 2026) established, in discrete games, that transition accuracy on randomly sampled play-throughs is the wrong adequacy criterion for a world model used in planning, and quantified the failure: danger = play_cost × $(1 - \text{rarity})^N$, with the gate-miss factor exact under i.i.d. sampling. Its related-work discussion closed with a promissory note: the rare-rule gap is a localized, discrete failure that state-accuracy metrics mask by dilution — a point worth revisiting in continuous settings, where the model-based RL literature (Lambert et al., 2020; Hafner et al., 2020; Janner et al., 2019) treats world-model error as pervasive and compounding rather than localized and pivotal.

This paper is that revisit. The question is not whether continuous world models can be wrong — that literature is mature — but whether the specific *verified-but-wrong* geometry of the discrete result exists in continuous state spaces: a model that passes a sampling gate cleanly, is (in a precise sense) almost-everywhere correct, and is still exploited catastrophically by the planner that trusts it.

The answer is yes, and the continuous version of the story is in one respect cleaner than the discrete one. The discrete headline had two components: (a) a provable one (when the rare rule is absent from the sample, no learner can infer it *from the sample* — an identifiability event with probability exactly $(1-r)^N$) and (b) an empirical one (the LLMs tested did not infer the rule *even when it was present* — "translation, not inference"). In the continuous *one-dimensional* instruments the second component **vanishes** for the models where we measured it: GPT-5.x infers an omitted hybrid mode from a handful of boundary-crossing transitions, writing the exact global rule. This turns out to be *geometry-dependent*: on a third instrument, a 4D bi-modal patch field whose modes are 2D regions rather than 1D clamps, the residual returns — neither GPT-5.x size recovers the region rule from data (0/156 mode-containing seeds pooled over the disc, a zero-curvature square ablation, and a strongest-effort prompting/budget treatment), and a Claude cross-family arm on the same cell reproduces the same failure classes (Section 7.1), so "translation, not inference" comes back on two-dimensional boundaries even for the model that erased it on the clamps. What remains provable is exactly the gate-miss and identifiability propositions (Section 3): with the mode absent from the sample — the $(1-r)^N$ event — no learner can recover it from the sample alone (a prior or the specification could still supply it; Proposition 4). Sample-covered exactness is the acceptance criterion itself; that every accepted artifact's *off-sample* content was also the true mode — with a single exception, a prior-injected phantom mode erring only where its sample is silent — is an empirical, code-inspected regularity for the models tested (GPT-5.x repairs every revealed clamp; Qwen stalls and the gate refuses its patches; Claude repairs most through a symmetry prior), not a theorem. The danger law here is not just exact; empirically, for the models tested, it is exhaustive *on the 1D instruments* — and Section 7.1 shows that this exhaustiveness is itself geometry-scoped, failing on a 2D circular mode.

Contributions:

1. **Theory transfer, and one new piece.** The gate-miss proposition, the identifiability proposition, and the play-cost upper bound via query-hit mass transfer to continuous state spaces with only notational change — they are measure-theoretic (Section 3) — and a corollary ties the bound to the paper's play_cost normalization (Corollary 1). The genuinely discrete ingredient of the companion paper is the *localization premise*, and we prove it requires unbounded local Lipschitz structure: for $L$-Lipschitz truth and model differing by $\eta$ at a state-action point, the disagreement region contains a metric ball of radius $(\eta-\varepsilon)/2L$ in $S \times A$ (Proposition 8). Hybrid mode boundaries are exactly where this obstruction disappears — an omitted mode is a rare rule.

2. **Two minimal hybrid instruments and two planner families** — cart-with-wall (linear off-mode plant) and pendulum-with-stop (nonlinear) — on which the full discrete phenomenology is measured with the same harness and no per-instrument re-calibration: the threshold law with the whole $(1-r)^N$ elbow inside the knob sweep, knob-invariant play_cost $\approx 1$ under random-shooting model-predictive control (MPC), and the reach mechanism in its cleanest form (Section 4). A fixed cross-entropy-method (CEM) planner occupies the bound's other branch: lower imagined mode-crossing and near-zero play cost on all 11 rows. Three instrument-design lessons of independent value are recorded (Section 2.3).

3. **A pinned-integrator contract and a measured axis separation.** The pinned-integrator synthesis contract makes correct code float-exact — a property of code exactness independent of the gate's tolerance — so the synthesis gate (Section 7) can run at $\varepsilon = 10^{-9}$. Separately, at a deployment-realistic $\varepsilon = 0.01$ (Section 5), the gate rejects supra-tolerance global bias on every rollout, passes sub-tolerance bias harmlessly, and misses the hard mode at a rate

that closely tracks the predicted $(1-r)^N$ (within sampling noise at gate scale; Section 5). A smooth localized perturbation of comparable rarity has $\approx 0$, at higher amplitude *negative*, play cost.

4. **A measured planner-side mitigation** (Section 6): distrust-region replanning — the planner checks its model's predictions against observed transitions and fences the refuted predictions — collapses the pinned-forever, below-random exploitation to a bounded first-contact transient on all 11 knob rows across both hand-written instruments, at zero cost when the model is right (bit-identical to plain MPC, tested). The fence exploits the hard-boundary structure of these modes (Section 6); the gate-side law is untouched: the gate still certifies a wrong model.

5. **The synthesis result: danger collapses to pure identifiability** (Section 7). Real-LLM arms (GPT-5.x, 20 seeds/cell) with the identifiability event logged per seed: mode absent $\Rightarrow$ 20/20 verified-blind-exploited (regret 0.999; Wilson lower bound 0.84 over two *disjoint* sample blocks, Section 7); mode present $\Rightarrow$ repaired to the exact global rule (both sizes 10/10, large in $\leq 1$ iteration), never a wrong artifact accepted in the GPT-5.x arms. Cross-family spot-checks in two more families reproduce the blind-exploited event in every family but diverge on repair — Qwen repairs none (superstitious patches the gate refuses), Claude (agent-relayed) repairs most through a symmetry prior that can certify an invented mode unfalsifiable by its gate sample — so repair-from-data is model-dependent in mechanism, while identifiability is a property of the sample, not the model. The whole result replicates on a second, nonlinear instrument (a pendulum with an angular hard stop): GPT-5.x repairs 62/62 mode-present seeds and every mode-absent seed is blind and exploited. On a third, 4D bi-modal instrument (PatchField2D) the repair finding *inverts*: neither GPT-5.x size recovers the circular mode from data (0/76 mode-containing seeds), reducing the disc to a half-plane, and a Claude cross-family arm fails the same way. Two ablations then pin the mechanism: a *zero-curvature square* is not repaired either (0/40 — with the inverse error, discs written on square evidence), and region-first guidance at $3\times$ budget eliminates the half-plane yet yields hull fits of the one-sided contact evidence (0/40) — **0/156 pooled**. Repair-from-data is geometry-dependent via a *template prior* over region forms, not curvature, and the law's empirical exhaustiveness is correspondingly scoped — while the all-or-nothing gate stays sound, certifying no partial artifact (Section 7.1).

6. **Smooth learners cannot localize** (Section 8): a closed-form linear fit on mode-free data passes both gates fully blind (identifiability is learner-independent); on mode-containing data it is tilted twelve orders of magnitude off-mode by four contact rows in 3200 while still missing the mode. Both the danger geometry and the repair capability are representational properties of code.

Scope up front: three instruments (two 1D — cart, pendulum — plus a 4D bi-modal PatchField2D), two base planner families (random-shooting MPC and CEM, one fixed configuration each), 20 seeds per headline synthesis cell across the instruments (both GPT-5.x sizes; plus the caught cell on the pendulum and two bi-knob cells on PatchField2D), a 3-seed cross-family spot-check per 1D instrument (not full sweeps), and a probe-grade multilayer perceptron (MLP). Section 10 gives the honest assessment and the full seed accounting.

## 2 The instruments

This section specifies the hand-written hybrid instruments and the planner that every later experiment reuses, and records the design lessons learned while calibrating them.

### 2.1 Cart-with-wall

State $(x, v)$; action $a \in [-a_{\max}, a_{\max}]$; semi-implicit Euler with fixed $dt$:

$$a \leftarrow \mathrm{clamp}(a), \qquad v' = v + (\mathrm{gain} \cdot a - \mathrm{drag} \cdot v)\, dt, \qquad x' = x + v'\, dt,$$

with defaults $dt = 0.1$, gain $= 3$, drag $= 0.3$, $a_{\max} = 1$. The **hybrid mode** is an inelastic wall at $x_{\mathrm{wall}}$: if $x' \geq x_{\mathrm{wall}}$, the next state is exactly $(x_{\mathrm{wall}}, 0)$. The **blind model** is the same code path with the wall branch removed — the hand-written on-manifold proxy for a CWM synthesized from a spec that omits the wall, *bit-exact off-mode by construction* (tested: exact float equality on wall-free trajectories). We call such a model **mode-blind** (the canonical term; instrument-specific synonyms like "wall-blind", or "blind" for short, mean the same thing). Reward is two sigmoid plateaus: a small reachable one on the left (0.3, at $x \leq -6$) and a large one on the right (1.0, at $x \geq 12$) whose approach every swept wall position blocks. Episodes are 80 steps from $x_0 \sim U(-0.5, 0.5)$, $v_0 = 0$.

The design requirements mirror the companion paper's material-at-cap instrument (paper 1's discrete instrument: a rare resource-cap rule likewise omittable from sampled play): (a) random rollouts rarely fire the mode — the wall position is the **rarity knob** ($r$ sweeps $0.317 \to 0.0024$ over $x_{\mathrm{wall}} \in [2, 10]$; Table 2); (b) the omission is *exploited*, not merely mispredicted — the wall-blind model predicts coasting through the wall toward the large plateau, so the planner it advises drives right and is pinned; (c) the truth planner's optimal play differs qualitatively — it goes left.

The second instrument (pendulum with a hard angular stop; nonlinear gravity term) is described with its results in Section 4.1, and the third (PatchField2D, a 4D bi-modal patch field) in Section 4.2.

### 2.2 Planner and play_cost

The planner is model-predictive control by random shooting (Nagabandi et al., 2018; Chua et al., 2018): at each step, sample candidate action sequences (piecewise-constant blocks plus the three constant sequences $\{-a_{\max}, 0, +a_{\max}\}$), roll each out on the *model*, take the best first action, replan. The planner is a deterministic function of its model's responses and the seed, so the play-cost upper bound via query-hit mass applies verbatim (Section 3; Corollary 1 ties that bound to the normalization below). Single-agent control makes play_cost a normalized regret, cleaner than a two-player arena (no opponent confound):

$$\mathrm{play\_cost} \;=\; \frac{J_{\mathrm{truth}} - J_{\mathrm{model}}}{J_{\mathrm{truth}} - J_{\mathrm{rand}}},$$

where $J_{\mathrm{truth}}$, $J_{\mathrm{model}}$, and $J_{\mathrm{rand}}$ are the returns of the truth-planner, the model-planner (written $J_{\mathrm{blind}}$ in the tables when the model is the blind one), and the uniform-random policy, all measured in the true environment on paired seeds. The blind planner can score *below random* (it is actively exploited), so the normalized value can exceed 1; we report it unclamped.

### 2.3 Three instrument-design lessons

Recorded because they will bite any continuous CWM study:

1. **I.i.d. per-step candidate sampling silently removes the mode from imagination.** With i.i.d. candidates, imagined displacement is diffusive; no sampled sequence reaches distant reward within the horizon, so the truth model and the blind model rank all candidates *identically* and the wall never enters imagination — the two arms become indistinguishable not because the models agree but because the planner never queries where they differ. Piecewise-constant blocks plus constant candidates fix it. The planner's *query* distribution, not just its trajectory distribution, is part of the instrument.

2. **Point (Gaussian) reward lodes demand braking finesse that random shooting lacks**; sigmoid plateaus remove the parking problem and give clean imagined-value margins in both directions.

3. **The plant's drag time-constant must sit well inside the planning horizon**, or no arm can act on the reward at all (our first calibration had $\tau = 1/\text{drag} = 10\,\text{s}$ against a $3\,\text{s}$ horizon; nothing moved).

# 3 Theory: what transfers, what changes, what is new

Notation: a verification gate draws $N$ i.i.d. rollouts from the gate policy $\rho$ (uniform-random actions from the initial-state distribution) and accepts a model $\hat{f}$ if it matches the truth $f$ within $\varepsilon$ in sup-norm on every visited transition.

**Measure-space setup.** Fix a horizon $T$. A trajectory is a point of $(S \times A)^T$, equipped with the product Borel $\sigma$-algebra. The gate policy $\rho$ together with the transition kernel induced by $f$ defines a trajectory law $P_\rho$ on this space (the initial-state distribution pushed forward through $\rho$ and $f$). The query measure $\mu_{\text{query}}(E)$ (the *query-hit mass*) is the probability, under the planner's trajectory law, that at least one model query during the episode lands in $E$. We assume throughout that the critical region $R$, the disagreement region $E$, and the reward and dynamics maps are Borel measurable, so every probability below is well defined. With this in place the discrete arguments of Aguilar Martín (2026) transfer with only notational change.

**Proposition 1** (gate miss; transfers verbatim)**.** *Let $R$ be any measurable set of rollouts ("the critical event"; here: the rollout fires the wall mode) with $r = P_\rho(R)$. The probability that $N$ i.i.d. gate rollouts all avoid $R$ is exactly $(1-r)^N$.*

Nothing in the companion paper's proof uses discreteness; the event is Bernoulli($r$) and the draws are i.i.d.

**Proposition 2** (joint gate miss: bracketed, not factored)**.** *Let $R_1, R_2$ be measurable critical events (two modes) with $r_i = P_\rho(R_i)$ and $r_\cup = P_\rho(R_1 \cup R_2)$. The probability that $N$ i.i.d. gate rollouts avoid* both *is exactly $(1-r_\cup)^N$, and with no independence assumption*

$$\big(1 - \min(1,\, r_1 + r_2)\big)^N \;\le\; (1 - r_\cup)^N \;\le\; \big(1 - \max(r_1, r_2)\big)^N,$$

*a bracket in which the product $\big((1-r_1)(1-r_2)\big)^N$ also lies. The product equals the truth iff $P_\rho(R_1 \cap R_2) = r_1 r_2$, and in general*

$$\frac{\big((1-r_1)(1-r_2)\big)^N}{(1-r_\cup)^N} = \left(1 + \frac{r_1 r_2 - P_\rho(R_1 \cap R_2)}{1 - r_\cup}\right)^N,$$

*so the product* over-*estimates the joint miss probability under negative dependence ($P_\rho(R_1 \cap R_2) < r_1 r_2$) and* under-*estimates it under positive dependence.*

*Proof.* Proposition 1 applied to the event $R_1 \cup R_2$ gives the exact value $(1-r_\cup)^N$. Monotonicity gives $r_\cup \geq \max(r_1, r_2)$ and subadditivity $r_\cup \leq \min(1, r_1 + r_2)$; since $x \mapsto x^N$ is increasing on $[0, 1]$, the bracket follows. For the product, $(1-r_1)(1-r_2) = 1-r_1-r_2+r_1r_2 \geq 1-r_1-r_2$, and $\leq 1-\max(r_1, r_2)$ because $r_1r_2 \leq \min(r_1, r_2)$. Finally inclusion–exclusion gives $1 - r_\cup = 1 - r_1 - r_2 + P_\rho(R_1 \cap R_2)$, hence $(1 - r_1)(1 - r_2) - (1 - r_\cup) = r_1r_2 - P_\rho(R_1 \cap R_2)$, which yields the displayed ratio and its sign. □

*Remark* 1 (the bracket is tight, and exact factorization is a property of the gate). The bracket is not merely valid but **sharp**: its ends are the Fréchet–Hoeffding bounds for $P_\rho(R_1 \cup R_2)$ given the marginals, pushed through the increasing map $x \mapsto x^N$, so no bound expressible in $r_1, r_2$ alone can be tighter — the interval is the *exact* range of joint miss probabilities consistent with the marginals. Both ends are attained: the lower one when the two critical events are disjoint ($P_\rho(R_1 \cap R_2) = 0$, whence $r_\cup = r_1 + r_2$), the upper one when one contains the other ($r_\cup = \max(r_1, r_2)$). This is the point of the bracket: it is the correct distribution-free statement about a multi-mode gate, replacing an independence assumption with no assumption at all, at the price of an interval instead of a point — and the interval cannot be narrowed without measuring something beyond the marginals (which is what $r_\cup$, or equivalently $P_\rho(R_1 \cap R_2)$, is). The lower end is attained in our data — at six of the nine PatchField2D knobs *no* rollout out of 600 contacts both patches, so the measured $r_\cup$ equals $r_1 + r_2$ exactly there (Table 4; a censored zero, so read it as $P(\text{both}) < 1/600$). Note that moving the modes apart pushes the joint factor *towards* that lower end rather than towards the product: disjointness is the opposite of independence, not a route to it. Exact factorization is instead a property of the *gate*: if it spends independent rollout budgets $N_1, N_2$ on the two mode regions — a *stratified* gate — the joint miss probability is exactly $(1 - r'_1)^{N_1}(1 - r'_2)^{N_2}$ in the per-stratum rarities, because the draws are then independent by construction. A multi-mode pipeline that wants a factorized guarantee has to stratify its gate; one undirected random budget only ever earns the bracket.

Section 4.2 measures both ingredients on the bi-modal instrument, and the proposition is confirmed where it is checkable: the bracket contains the measured joint factor at all nine knobs, and the sign rule predicts the direction of the product's error at all nine — an error that runs $-17\%$ to $+12\%$ because the dependence changes sign across the grid. The practical reading is that a multi-mode danger law needs the union event measured (or the bracket, which needs only the marginals), never the marginals multiplied.

**Proposition 3** ($\varepsilon$-invariance of a hard mode's reveal-rarity, with its exact threshold)**.** *Fix a hard mode, a gate policy, and the mode-blind model. For a rollout $\omega$ put*

$$D(\omega) \;=\; \max\{\, \|f(s,a) - \hat{f}(s,a)\|_\infty \;:\; (s,a) \text{ a mode contact of } \omega \,\},$$

*with $D(\omega) = 0$ if $\omega$ never contacts the mode. Then the reveal-rarity at tolerance $\varepsilon$ — the probability that a gate rollout contains a transition on which truth and model differ by more than $\varepsilon$ — is exactly $P_\rho(D > \varepsilon)$, a non-increasing step function of $\varepsilon$ whose steps sit at the realized values of $D$. Consequently, with $\varepsilon^* = \min\{D(\omega) : D(\omega) > 0\}$,*

$$\textit{reveal-rarity}(\varepsilon) \;=\; \underbrace{P_\rho(D > 0)}_{\textit{mode-firing rarity } r} \qquad \textit{for every } \varepsilon < \varepsilon^*,$$

*and* pass@$N = (1 - r)^N$ *is* exactly *$\varepsilon$-invariant on that range, degrading only above $\varepsilon^*$. So writing one symbol $r$ for the two rarities is licensed precisely for $\varepsilon < \varepsilon^*$, and the flatness the sweep measures is an identity there rather than a coincidence.*

Note what $\varepsilon^*$ is: a per-rollout *minimum* of a per-contact *maximum*. A pinned rollout contacts the mode many times, and it only has to contact it coarsely *once* to be revealed, which is why $\varepsilon^*$ sits two orders of magnitude above the smallest single-contact disagreement (0.0017 on `wall@4`). The threshold is computable (`scripts/eps_invariance_threshold.py`), and it **predicts where each sweep arm dips**, arm by arm:

| mode arm | $\varepsilon^*$ | first grid $\varepsilon \geq \varepsilon^*$ | measured in the sweep |
|---|---|---|---|
| cart `wall@8` | 0.3855 | none in grid | bit-identically flat through $\varepsilon = 0.3$ |
| cart `wall@4` | 0.0561 | 0.1 | flat to $3\times10^{-2}$, slight dip from 0.1 |
| pendulum `stop@1.0` | 0.0513 | 0.1 | 0.1410 to $3\times10^{-2}$, 0.1400 at 0.1, 0.1240 at 0.3 |
| pendulum `stop@1.4` | 0.1164 | 0.3 | flat to 0.1, dips at 0.3 |

Table 1: The $\varepsilon$-invariance threshold predicts the sweep. $\varepsilon^*$ is computed from the gate policy alone (2000 rollouts/arm, no gate involved); the last column is what Section 5's sweep measured independently. Every arm's first departure from flatness lands at the predicted grid point — including `wall@8`, whose $\varepsilon^*$ exceeds the grid and which is therefore flat throughout.

**Proposition 4** (identifiability; transfers verbatim)**.** *Let $M \subseteq S \times A$ be the mode region (for the cart the clamp fires depending on $(s, a)$ through $x' \geq x_{\text{wall}}$) and instantiate Proposition 1 at the critical event $R = \{$rollouts that visit $M\}$, so that "miss" means exactly that no sampled transition lies in $M$. Condition on that miss event and let $\hat{f}_1, \hat{f}_2$ be any two models that agree on $(S \times A) \setminus M$. On the miss event the sample visits no transition in $M$, so $\hat{f}_1$ and $\hat{f}_2$ produce identical outputs on every sampled input; hence every sample-measurable score (the gate, a likelihood, a refinement objective) is constant across the two, and no such score can distinguish them — preference for the correct one must come from the prior or the specification. This holds for* any *learner — LLM, linear regression, MLP — a point Section 8 instantiates empirically.*

**Proposition 5** (play-cost upper bound; transfers verbatim)**.** *Assume returns are normalized to* $[0, 1]$ *(WLOG, by rescaling $J \mapsto (J - J_{\min})/(J_{\max} - J_{\min})$; equivalently carry an explicit factor* $J_{\max} - J_{\min}$ *throughout). For any planner that is a deterministic function of model responses and a seed, $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$, where $E$ is the disagreement region and $\mu_{\text{query}}$ the probability (the query-hit mass) that the planner queries its model on $E$ during an episode.*

*Proof sketch (coupling).* Couple the two runs on a common seed. On the event that no query lands in $E$, $f$ and $\hat{f}$ return identical responses to every query the planner issues (they agree off $E$), so with shared rng the two runs select the same action sequence and traverse the same states, realizing the same return. That event is unambiguous: the two runs are identical up to the first query landing in $E$, so "no query ever lands in $E$" is a single coupling-measurable event with one probability, not two model-dependent ones. The realized returns can therefore differ only on its complement, of probability $\mu_{\text{query}}(E)$, where normalized returns differ by at most 1; taking expectations gives $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$. MPC's imagined rollouts are the queries. (The argument is the coupling of Aguilar Martín (2026), recorded here rather than deferred.)

The raw $J$ reported in the tables (e.g. $J_{\text{truth}} = 17.77$) is this normalized quantity rescaled by $J_{\max} - J_{\min}$, so the bound is a statement about the normalized return. The paper's reported play_cost, however, divides by $J_{\text{truth}} - J_{\text{rand}}$ (Section 2.2), *not* by $J_{\max} - J_{\min}$; the following corollary states the relationship rather than silently identifying the two.

**Corollary 1** (play_cost saturation)**.** *Let $J_{\max}, J_{\min}$ be the extremes of the* realizable *return on the instrument (the supremum and infimum over policies, not over arbitrary state sequences) — the* *reading under which Proposition 5's normalization is not vacuous. With* $|\text{play\_cost}| = |J(f) - J(\hat{f})|/(J_{\text{truth}} - J_{\text{rand}})$*, play_cost as defined (signed) in Section 2.2, Proposition 5 in raw units* ($|J(f) - J(\hat{f})| \le \mu_{\text{query}}(E)\,(J_{\max} - J_{\min})$) *gives*

$$|\text{play\_cost}| \;\le\; \mu_{\text{query}}(E)\,\frac{J_{\max} - J_{\min}}{J_{\text{truth}} - J_{\text{rand}}},$$

*so the bound reads* $|\text{play\_cost}| \lesssim \mu_{\text{query}}(E)$ *exactly when* $J_{\text{rand}} \approx J_{\min}$ *and* $J_{\text{truth}} \approx J_{\max}$ *— the random policy near the reward floor, the truth planner near the ceiling. Both instruments are in this regime: on the cart* $J_{\text{rand}} = 0.53$ *against a pinned* $J_{\text{blind}} \approx 0$ *and* $J_{\text{truth}} = 17.77$*; on the pendulum* $J_{\text{rand}} = 0.06$ *and* $J_{\text{truth}} = 20.08$*. The blind planner queries the wall region in every episode, so* $\mu_{\text{query}}(E) \approx 1$ *and the normalized bound is saturated —* play_cost $\approx 1$*, as observed in Sections 4–5. On the cart the bound is in fact* attained*: with* $\mu_{\text{query}} = 1$*,* $J_{\max} = J_{\text{truth}}$ *and the blind planner pinned at the realizable floor* $J_{\min} = 0$*, bound and measurement coincide at* $J_{\text{truth}}/(J_{\text{truth}} - J_{\text{rand}}) = 17.77/17.24 = 1.031$ *(Table 2) — attained* because *the exploited planner reaches the floor, not by coincidence.*

**Proposition 6** (knob-invariance of play_cost is arithmetic, not a regularity)**.** *Let the rarity knob* $k$ *range over values for which the truth planner's realized return* $J_{\text{truth}}$ *and the random policy's return* $J_{\text{rand}}$ *do not depend on* $k$*. Then, directly from the definition of Section 2.2,*

$$\text{play\_cost}(k) \;=\; \underbrace{\frac{J_{\text{truth}}}{J_{\text{truth}} - J_{\text{rand}}}}_{\textit{knob-free}} - \frac{J_{\text{blind}}(k)}{J_{\text{truth}} - J_{\text{rand}}},$$

*an affine function of the exploited planner's own return with knob-independent coefficients. Hence play_cost is constant up to*

$$\max_k \text{play\_cost}(k) - \min_k \text{play\_cost}(k) \;=\; \frac{\max_k J_{\text{blind}}(k) - \min_k J_{\text{blind}}(k)}{J_{\text{truth}} - J_{\text{rand}}},$$

*so the entire knob-dependence is the exploited planner's* residual *reward divided by the truth-minus-random margin; a planner pinned with none gives play_cost exactly* $J_{\text{truth}}/(J_{\text{truth}} - J_{\text{rand}})$*.*

The two hypotheses are measured, and measured sharply: on the cart the truth planner's return is knob-independent *to twelve digits* ($J_{\text{truth}} = 17.757356407381$ at all seven knobs of the sharp variant, 17.772246981024 at all seven of the default), and $J_{\text{rand}}$ varies across knobs by $3.9 \times 10^{-9}$ on the sharp variant and $1.2 \times 10^{-4}$ on the default — the random policy does occasionally enter the mode region, so its knob-independence is approximate rather than structural, and measurably less approximate once the plateau tail is removed. With those in hand the proposition is not decoration: it predicts every measured play_cost to $2.4 \times 10^{-10}$ on the sharp variant, and that residue is itself accounted for rather than left over: $\partial\,\text{play\_cost}/\partial J_{\text{rand}} = (J_{\text{truth}} - J_{\text{blind}})/(J_{\text{truth}} - J_{\text{rand}})^2 \approx 0.060$, which against $J_{\text{rand}}$'s measured $3.9 \times 10^{-9}$ knob-variation predicts $2.3 \times 10^{-10}$ — the observed residue to one digit. On the default the same propagation of a $1.2 \times 10^{-4}$ variation gives $7 \times 10^{-6}$, matching the $7.3 \times 10^{-6}$ observed there, and the spread it predicts, $1.3615 \times 10^{-4}$, matches the measured $1.3615 \times 10^{-4}$. So the "knob-invariant play_cost $\approx 1$" of Section 4 is an *identity* once the planner is pinned, and what the sharp-plateau variant buys is not invariance itself but the vanishing of $J_{\text{blind}}$ that makes the identity's residual term negligible. What remains empirical is the pinning (contact = 1.00 and $J_{\text{blind}} \approx 0$) and the truth planner's indifference to the knob. The latter is not automatic — the clamp does change the imagined value of right-going candidates — so we do not leave it at bit-identity. A candidate's imagined return under truth depends on the knob *only if* its imagined trajectory reaches the clamp, so it suffices to check, at every knob and every replanning step, that (C1) the argmax candidate's imagined trajectory stays strictly below the smallest wall in the sweep, and (C2) every candidate that does reach the wall scores strictly below it. Then the argmax is the maximiser over a knob-free subset and every knob-dependent candidate loses, so the chosen action, the realized trajectory and $J_{\text{truth}}$ are identical across knobs — a machine-checked certificate over the planner's own candidate set rather than an observation about outputs. `scripts/truth_plan_invariance_certificate.py` runs it: the certificate holds at *every* knob of the sweep, with the argmax candidate never exceeding $x = 0.344$ and clamping candidates losing by a margin of 5.25 or more. Pushed past the sweep, the margin contracts monotonically — 4.12 at $x_{\text{wall}} = 11$, 0.51 at 12 — and the certificate fails at 12.5, where the argmax candidate itself reaches the wall. So the regime has a *derived* boundary at $x_{\text{wall}} \approx x_{\text{right}} = 12$: the truth planner switches to the right plateau exactly once the wall stops blocking it, which is also where the mode stops being a trap and there is no danger left to measure. The sweep $[2, 10]$ sits inside the certified region with a wide margin.

**What does not transfer.** The companion paper's coverage certificates enumerate finite information-set spaces. Continuous state spaces admit no such enumeration; covering-number analogues under Lipschitz assumptions are left open (Section 10).

**What the gate *does* certify: the continuous coverage analogue.** The companion paper's coverage guarantee enumerates information sets, which is why its transfer was listed as open. With the disagreement ball of Proposition 8 and the visitation density of Remark 2 the analogue is available, with a covering number in place of the enumeration.

**Proposition 7** (coverage certificate for Lipschitz models)**.** *Let $f, \hat{f}$ be $L$-Lipschitz on $U \subseteq S \times A$ in the sup-metric, and suppose $\hat{f}$ passes the gate at tolerance $\varepsilon$ on every visited transition. If the visited set is a $\rho$-net of $U$, then*

$$\sup_{u \in U} \|f(u) - \hat{f}(u)\|_\infty \;\leq\; \varepsilon + 2L\rho.$$

*Moreover, if the gate's per-step visitation density is at least $c$ on $U$ and $M$ independent samples are drawn, then the visited set is a $\rho$-net with probability at least $1 - \delta$ as soon as*

$$M \;\geq\; \frac{\ln\big(N_{\text{cov}}(U, \rho/2)/\delta\big)}{c \cdot \text{vol}(B_{\rho/2})},$$

*where $N_{\text{cov}}(U, \rho/2) \leq \text{vol}(U \oplus B_{\rho/4})/\text{vol}(B_{\rho/4})$.*

*Proof.* For the first part, take $u \in U$ and let $v$ be a visited point with $\|u - v\|_\infty \leq \rho$; then $\|f(u) - \hat{f}(u)\| \leq \|f(v) - \hat{f}(v)\| + \|f(u) - f(v)\| + \|\hat{f}(u) - \hat{f}(v)\| \leq \varepsilon + 2L\rho$. For the second, fix a maximal $(\rho/2)$-packing of $U$; it is a $(\rho/2)$-cover, its cardinality is bounded as stated (the disjoint $(\rho/4)$-balls sit inside $U \oplus B_{\rho/4}$), and each packing point's $(\rho/2)$-ball has probability at least $c\,\text{vol}(B_{\rho/2})$. The chance that some such ball receives no sample is at most $N_{\text{cov}}(1-p)^M \leq N_{\text{cov}} e^{-pM}$, which is $\leq \delta$ under the displayed condition; when every ball is hit, each $u \in U$ lies within $\rho/2$ of a packing point that holds a visited sample, hence within $\rho$ of a visited point. □

Instantiated on the deployed cart gate (`scripts/gate_coverage_certificate.py`; $U$ the step-1 reachable set of Corollary 2, $c = 5/6$, $L = 1.27$, $\varepsilon = 0.01$, $\delta = 0.05$): one step per rollout — the reading that needs no assumption about steps within a rollout — gives $\rho = 0.615$ from $N = 40$ rollouts, certifying $\sup_U \|f - \hat{f}\|_\infty \leq 1.57$.

**Using all the steps, without assuming they are independent.** The obvious objection is that the gate takes $40 \times 80$ steps and the above uses 40 of them. The dependence can be handled exactly rather than assumed away, because of a structure the gate policy has: the action $a_t$ is drawn i.i.d. and independently of $s_t$. Conditioning on a rollout's *entire* state trajectory, the action indicators at the visiting times are independent Bernoulli($q_a$), so for a product cell $C = C_s \times C_a$

$$P(C \text{ unhit by a rollout} \mid \text{state trajectory}) = (1 - q_a)^O, \qquad O = \#\{t : s_t \in C_s\},$$

which is exact and uses no independence between steps; and since *rollouts* are i.i.d. the expectation factorizes, $P(C \text{ unhit by the gate}) = \varphi^N$ with $\varphi = \mathbb{E}[(1-q_a)^O]$ a per-rollout expectation of a $[0,1]$ variable, hence Monte-Carlo boundable. Requiring $\varphi^N \le \delta/2K$ per cell (`scripts/gate_coverage_dependent.py`, 8000 MC rollouts, Hoeffding on $\varphi$) the deployed gate certifies $\rho = 0.60$ and $\le$ **1.53**, needing $N \ge 35$ against the 40 it has. *So the mixing question was a red herring*: handling the dependence exactly buys 2% ($1.57 \to 1.53$), not the factor of four that treating all 3200 steps as independent would suggest. The binding constraint is not how steps correlate but how many *rollouts* reach the worst cell, and the resulting gate-size/resolution trade-off is itself the quantitative statement of a sampling gate's weakness: $\rho = 0.5$ needs $N \ge 169$, $\rho = 0.4$ needs $N \ge 1179$, i.e. certifying a bound of 1.03 instead of 1.53 costs a gate thirty times larger.

**The density at every step, and what it buys.** The certificate above lives on the one-step reachable set because that is where Corollary 2 derives the density. The restriction lifts: the plant is linear and time-invariant, so $(x_t, v_t)$ is an affine image of $(x_0, a_0, \ldots, a_{t-1})$, and splitting off the *last two* actions gives $(x_t, v_t) = W_t + M(a_{t-2}, a_{t-1})^\top$ with $M$ the (time-invariant) Jacobian

$$M = \begin{pmatrix} \text{gain}\, dt^2 & \text{gain}\, dt^2(1 + (1 - \text{drag}\, dt)) \\ \text{gain}\, dt & (1 - \text{drag}\, dt)\, \text{gain}\, dt \end{pmatrix}, \qquad |\det M| = 0.009$$

(verified against finite differences). So at *every* $t \ge 2$ the step-$t$ law is a convolution with the uniform law on a parallelogram of area $4a_{\max}^2 |\det M| = 0.036$, hence absolutely continuous with

$$p_t(u) \;\ge\; 27.78 \cdot P\big(M^{-1}(u - W_t) \in [-a_{\max}, a_{\max}]^2\big),$$

an exact constant times a reachability probability. Two lessons from instantiating it (`scripts/gate_density_step_t.py`). First, the certified region must be a *level set* of the density and not a box: a box's corners pair extreme $x$ with extreme $v$, which the gate does not reach, so an infimum over a box is 0 and the bound is vacuous — we measured that before believing it. Second, later steps trade density for extent: at step 20 the level set $\{p \ge 0.05\}$ has volume 6.9 in $(x, v, a)$ against the one-step set's 1.2, and $N = 40$ then certifies $\rho = 1.44$, i.e. $\sup \|f - \hat{f}\| \le 3.67$ — **a region** 5.7× **larger, still excluding the hard mode's** 4.2. Pushing further (step 40, $\{p \ge 0.01\}$, volume 19) gives $\rho = 2.39$ and a bound of 6.07, which no longer excludes it: that is the frontier of what 40 rollouts can certify, and it is a statement about the gate, not about the technique.

**The nonlinear case is not a separate case: the constant is universal.** The step-$t$ derivation above used the cart's linearity, and the natural worry is that a nonlinear plant needs its own argument. It does not. For *any* plant of the semi-implicit form

$$\omega_{t+1} = \omega_t + \big(\text{gain} \cdot a_t + F(\theta_t, \omega_t)\big)\, dt, \qquad \theta_{t+1} = \theta_t + \omega_{t+1}\, dt,$$

with $F \in C^1$ arbitrary, the Jacobian of the last two actions satisfies

$$\big|\det \partial(\theta_t, \omega_t)/\partial(a_{t-2}, a_{t-1})\big| \;=\; \text{gain}^2\, dt^3 \quad \text{identically,}$$

independent of $F$, of the state and of $t$. Writing $F_\theta, F_\omega$ for the partials: $\partial\omega_t/\partial a_{t-1} = \mathrm{gain}\, dt$ and $\partial\theta_t/\partial a_{t-1} = \mathrm{gain}\, dt^2$; $\partial\omega_t/\partial a_{t-2} = \mathrm{gain}\, dt(1 + F_\omega dt) + F_\theta\, \mathrm{gain}\, dt^3$ and $\partial\theta_t/\partial a_{t-2} = \mathrm{gain}\, dt^2 + dt\, \partial\omega_t/\partial a_{t-2}$; substituting, every $F$-dependent term cancels and the determinant is $-\mathrm{gain}^2 dt^3$. Checked numerically against four forces including a non-separable one, at several $(\mathrm{gain}, dt)$, to ten digits. So the density constant $c = 1/(4a_{\max}^2 \mathrm{gain}^2 dt^3) = 27.78$ is the *same* for the cart and the pendulum, and the certificate's hypothesis holds for the whole instrument family — with one exception, and it is the expected one: wherever the mode clamp fires, the two-action map is constant, so det $= 0$ exactly and no density argument reaches the mode. The obstruction is intrinsic, not a limitation of this technique.

**But the certified region is not where the planner looks.** The certificate bounds the model on the region the *gate* covers; the play cost is driven by the region the *planner* queries (Proposition 5). Those are different measures, and the honest way to relate them is to measure the overlap (`scripts/certified_region_query_mass.py`: every imagined MPC step is a query). Of the exploited planner's 1.3 million queries per episode pair, 1.9% fall inside the region certified at $\rho = 0.60$ and 7.8% inside the larger step-20 region; for the truth planner, 2.3% and 7.3%. So the certificate is sound and *nearly irrelevant to play*: not because the bound is loose — it is tight to one grid step — but because the gate covers a couple of percent of the query mass. That is this paper's thesis restated as a measurement rather than an argument, and it is the reason the continuous coverage analogue, once obtained, does not rescue sampling verification: *the gate certifies where it looks, and the planner looks somewhere else.*

**Is the bound tight?** A bound that merely holds could be loose by orders of magnitude, so we measured the truth (`scripts/gate_coverage_validation.py`, 200 independent gates per resolution): the deployed gate actually $\rho$-covers in 200/200 trials at $\rho = 0.70$, 199/200 at $\rho = 0.60$ — the resolution the certificate licenses — then 128/200 at 0.50 and 10/200 at 0.40, exactly where the certificate says a much larger gate is needed ($N \geq 169$ and $N \geq 1179$). The threshold is therefore tight to one grid step, which is what makes the $N$-versus-$\rho$ trade-off worth quoting.

Both figures matter less than what they exclude: the hard mode's own disagreement is 4.2 (the wall-region probe error of Section 8), *above both bounds*, so **no Lipschitz model with the wall's error magnitude can pass this gate** — and the wall does pass it, escaping the certificate by the only route left, namely by not being Lipschitz. The continuous coverage analogue therefore closes exactly the case the paper says it closes (smooth models) and is silent exactly where the danger lives (hybrid modes), which is the same boundary Proposition 8 draws.

**What is new: the localization premise is a theorem-shaped obstruction.**

**Proposition 8** (smoothness forbids localized error)**.** *Let $f, \hat{f} : S \times A \subseteq \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$ be $L$-Lipschitz on the joint state-action space in the sup-metric on $S \times A$, with $L \in [0, \infty)$ (i.e. $L = \max(\mathrm{Lip}\, f, \mathrm{Lip}\, \hat{f})$). Suppose $\|f(s_0, a_0) - \hat{f}(s_0, a_0)\|_\infty = \eta$ at some $(s_0, a_0) \in S \times A$. Fix any tolerance $\varepsilon < \eta$ and write $E_\varepsilon = \{(s, a) \in S \times A : \|f(s, a) - \hat{f}(s, a)\|_\infty > \varepsilon\}$. If $L > 0$, then $E_\varepsilon$ contains the open metric ball $B\big((s_0, a_0), \frac{\eta-\varepsilon}{2L}\big) \cap (S \times A)$ (the intersection matters only when $(s_0, a_0)$ lies near $\partial(S \times A)$). If $L = 0$, then $f - \hat{f}$ is constant and $E_\varepsilon$ is all of $S \times A$ — the degenerate extreme of the same statement, and the reason the radius is written for $L > 0$.*

*Proof.* $g = f - \hat{f}$ is $2L$-Lipschitz on $S \times A$, so for $\|(s, a) - (s_0, a_0)\|_\infty < (\eta - \varepsilon)/2L$ we have $\|g(s, a)\|_\infty \geq \eta - 2L\, \|(s, a) - (s_0, a_0)\|_\infty > \varepsilon$; the inequality is strict, so the ball is open. The $L = 0$ case is immediate: $g \equiv g(s_0, a_0)$ with $\|g(s_0, a_0)\|_\infty = \eta > \varepsilon$. $\square$

The disagreement region $E_\varepsilon$ lives in the same joint state-action space $S \times A$ as the query region $E$ of Proposition 5, so the ball is *directly* a lower bound on the query-relevant region the planner can hit — no fixed-action slice argument is needed to bring the two together. (Restricting to a fixed action $a_0$ recovers the state-slice form: the open ball of radius $(\eta - \varepsilon)/2L$ in $S \times \{a_0\}$ lies in $E_\varepsilon$, the version convenient when reasoning about a single instrument's mode.)

Read as its contrapositive: a model error that is *large somewhere* (large enough to matter at play) and *invisible at tolerance $\varepsilon$ outside a metrically tiny region* requires $L$ large — in the limit, a discontinuity. This is exactly the sense in which the discrete localization premise ("the wrong model is exact off the rule region") is discrete: it is unsatisfiable by smooth truth/model pairs, up to the ball-radius quantification. Two consequences structure the paper: (i) the natural continuous home of the danger geometry is **hybrid dynamics** — contacts, hard stops, saturations, regime switches — where the truth itself has unbounded local Lipschitz constant across a mode boundary (our wall: $v$ jumps to 0), and an omitted mode is precisely a rare rule; (ii) the premise is *representationally* available to programs (an omitted or added branch is bit-exact off the omitted case) and not to smooth learned models, a distinction Section 8 measures.

**Proposition 9** (smooth localized error is detectable at a rate)**.** *In the setting of Proposition 8 with $L > 0$, let $\rho = (\eta - \varepsilon)/2L$ be the radius of the guaranteed disagreement ball $B \subseteq E_\varepsilon$. Fix a step index $k \leq T$ and suppose the gate's step-$k$ visitation law admits a density bounded below by $c > 0$ with respect to Lebesgue measure on $B$. Then one gate rollout reveals the disagreement with probability at least*

$$q \;=\; c\,(2\rho)^{d+m} \;=\; c\left(\frac{\eta-\varepsilon}{L}\right)^{d+m}$$

*(the sup-metric ball is a cube of side $2\rho$), so by Proposition 1 the gate's miss probability is at most $(1-q)^N$. Equivalently — what it takes to* hide *— keeping the miss probability above $\delta$ against $N$ rollouts forces*

$$L \;\geq\; (\eta-\varepsilon)\left(\frac{c\,N}{\ln(1/\delta)}\right)^{1/(d+m)}.$$

*Proof.* $B \subseteq E_\varepsilon$ by Proposition 8, and the step-$k$ visitation of $B$ has probability at least $c \cdot \mathrm{vol}(B) = c(2\rho)^{d+m}$ by the density hypothesis; a rollout that visits $E_\varepsilon$ at step $k$ reveals a disagreement, so the per-rollout reveal probability is at least $q$ and the critical event of Proposition 1 has $r \geq q$, giving $P(\text{miss}) = (1-r)^N \leq (1-q)^N$. For the converse, $(1-q)^N > \delta$ requires $q < 1 - \delta^{1/N} \leq \ln(1/\delta)/N$; substituting $q = c((\eta-\varepsilon)/L)^{d+m}$ and solving for $L$ gives the display. □

Three readings. (i) It is the *measure* counterpart of Proposition 8: smoothness does not merely forbid exact localization, it forces a detection rate. (ii) The Lipschitz constant needed to hide a fixed error grows only like $N^{1/(d+m)}$, so at fixed $c$ hiding gets *easier* with dimension — the same dimensional loosening paper 3 studies as a rarity knob. (At fixed $c$ is the honest qualifier: $c$ carries dimensions of inverse volume, so the dimension-free way to say it is Remark 2's volume ratio — the ball's share of the reachable set shrinks with dimension at fixed radius.) (iii) The hypothesis is where the work is: verifying a visitation-density lower bound is instrument-specific, and our gate's step-1 law is supported on a lower-dimensional set (the initial-state distribution has $v = 0$), so we do not claim $c$ for the instruments here. What we *do* have is the direct measurement in its place: the smooth bump arm of Section 5 has reveal-rarity 0.19 against the hard wall's 0.14 at comparable amplitude — the smooth error is, if anything, *more* detectable, exactly as this proposition's direction requires, and its harmlessness comes from play_cost $\approx 0$ rather than from hiding.

*Remark* 2 (the constant in general: a volume ratio). Proposition 9's $c$ is not an instrument-specific fudge; it is the gate's step-$k$ visitation density, which for a gate policy with box-uniform initial states and actions is available in closed form by change of variables. If the step-$k$ law of $(s, a)$ is uniform on a set $U \subseteq S \times A$ — as it is at $k = 1$ whenever the one-step map is affine in the randomized coordinates — then $c = 1/\mathrm{vol}(U)$ exactly, and the proposition's conclusion reads

$$q \;\geq\; \frac{\mathrm{vol}(B)}{\mathrm{vol}(U)},$$

*the fraction of the gate's one-step reachable volume that the guaranteed disagreement ball occupies.* For non-uniform laws the same computation gives $c = \inf_B p_0 / \sup_B |\det DF|$ through the push-forward, with $F$ the one-step map; the uniform case is the one where the two infima are attained everywhere. The dimensional reading of Proposition 9 is then geometric rather than analytic: hiding is easy exactly when the ball is a small fraction of the reachable volume, and raising $d + m$ shrinks that fraction for fixed radius.

**Corollary 2** (the constant, computed for this instrument)**.** *For the cart's gate at step $k = 1$ the density hypothesis is not an assumption: the gate policy gives $v_1 = \mathrm{gain} \cdot dt \cdot a_0$ with $a_0 \sim U(-a_{\max}, a_{\max})$, $x_1 = x_0 + dt\, v_1$ with $x_0 \sim U(-\frac{1}{2}, \frac{1}{2})$, and $a_1 \sim U(-a_{\max}, a_{\max})$ independent, so on $\{|v_1| < \mathrm{gain}\, dt\, a_{\max}\} \cap \{|x_1 - dt\, v_1| < \frac{1}{2}\} \cap \{|a_1| < a_{\max}\}$ the joint $(x_1, v_1, a_1)$ density factorizes exactly:*

$$c \;=\; \frac{1}{2\,\mathrm{gain}\, dt\, a_{\max}} \cdot 1 \cdot \frac{1}{2a_{\max}} \;=\; \frac{5}{6} \qquad \text{at } \mathrm{gain} = 3,\; dt = 0.1,\; a_{\max} = 1$$

*($d + m = 3$ here). Equivalently, by Remark 2, the step-1 law is uniform on a set of volume* $\mathrm{vol}(U) = (2\,\mathrm{gain}\, dt\, a_{\max}) \cdot 1 \cdot (2a_{\max}) = 1.2$ *and* $c = 1/1.2 = 5/6$ *— the reveal probability is literally the ball's share of the one-step reachable volume. Monte Carlo on an interior box agrees to* 1.3% *(`scripts/gate_density_constant.py`). Proposition 9 is therefore quantitative on this instrument: hiding an $\eta$-sized disagreement from the deployed gate ($\varepsilon = 0.01$, $N = 40$) with probability above $\frac{1}{2}$ requires*

$$L \;\geq\; (\eta - \varepsilon) \left( \frac{5/6}{\ln 2/40} \right)^{1/3},$$

*i.e. $L \geq 0.33$ for $\eta = 0.1$, $L \geq 1.78$ for $\eta = 0.5$, and $L \geq 15.2$ for $\eta = 4.2$ — against the plant's own sup-metric Lipschitz constant of* 1.27*. The last row is the instrument's own numbers:* 4.2 *is the wall-region probe error of Section 8 (a model that predicts straight through the wall), so* ***no smooth pair less than twelve times as sensitive as the plant can hide the wall from this gate****. That is the quantitative form of Proposition 8's obstruction, and it is why the smooth arms of Section 5 are* detected *(bump reveal-rarity* 0.18*) rather than hidden — their harmlessness comes from play_cost, not from invisibility.*

*Remark* 3 (metric, not measure). The ball in Proposition 8 is metric, and its *probability* under the gate's visitation measure can still be small — smoothness bounds how spatially concentrated an error can be, not how often the gate visits it. The proposition removes the *exact* localization premise for smooth pairs; the $(1 - r)^N$ mechanism itself is representation-independent and applies to any critical region of small visitation measure.

# 4 The mechanism and the threshold law

This section measures the danger law's full phenomenology on the hand-written instruments, before any LLM enters the loop. All numbers CPU-only, with the hand-written blind model as the on-manifold proxy (`scripts/continuous_reach.py`; 3000 rarity rollouts and 20 MPC episodes/arm per knob; Wilson 95% CIs (Wilson, 1927) in the versioned results).

| $x_{\text{wall}}$ | rarity | $J_{\text{truth}}$ | $J_{\text{blind}}$ | $J_{\text{rand}}$ | play_cost | blind hit | truth hit | d@20 | d@40 | d@80 |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 0.3168 | 17.77 | 0.00 | 0.53 | 1.031 | 1.00 | 0.00 | 0.001 | 0.000 | 0.000 |
| 3 | 0.2135 | 17.77 | 0.00 | 0.53 | 1.031 | 1.00 | 0.00 | 0.008 | 0.000 | 0.000 |
| 4 | 0.1352 | 17.77 | 0.00 | 0.53 | 1.031 | 1.00 | 0.00 | 0.056 | 0.003 | 0.000 |
| 5 | 0.0797 | 17.77 | 0.00 | 0.53 | 1.031 | 1.00 | 0.00 | 0.196 | 0.037 | 0.001 |
| 6 | 0.0439 | 17.77 | 0.00 | 0.53 | 1.031 | 1.00 | 0.00 | 0.420 | 0.171 | 0.028 |
| 8 | 0.0114 | 17.77 | 0.02 | 0.53 | 1.030 | 1.00 | 0.00 | 0.818 | 0.650 | 0.410 |
| 10 | 0.0024 | 17.77 | 0.94 | 0.53 | 0.977 | 1.00 | 0.00 | 0.930 | 0.886 | 0.804 |

Table 2: The danger curve on the wall-position knob. "blind/truth hit" = fraction of episodes in which that planner's trajectory fires the wall mode; d@$N$ = play_cost $\cdot (1-r)^N$. Rarity is measured on **30,000** random rollouts (Wilson 95% CIs in the versioned JSON), the same sample size as the pendulum's, so the mode-firing and reveal-rarity estimates of the same event are now directly comparable.

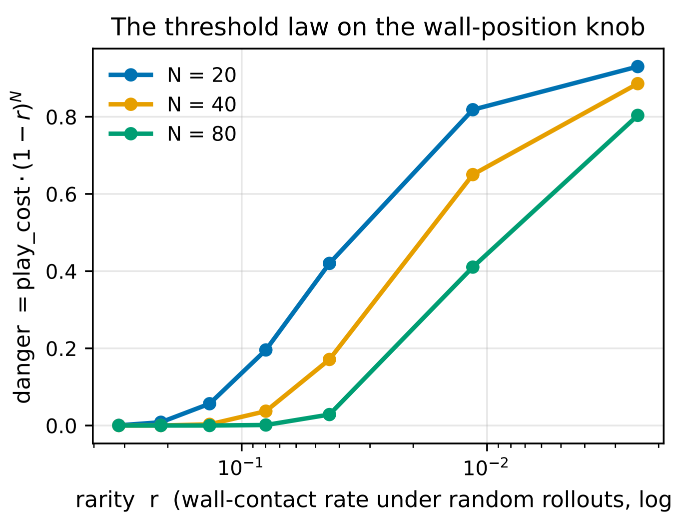


Figure 1: The threshold law on the wall-position knob (Table 2 rendered): danger $\approx 0$ while the wall is inside the random envelope, rises through the elbow, plateaus at full play_cost; $N$ shifts the threshold.

Readings: **the threshold law again** — danger $\approx 0$ while the wall sits inside the random-rollout envelope, rises through the elbow, and plateaus at full play_cost, with the entire elbow inside the sweep (Figure 1). **play_cost is knob-invariant and** $\approx 1$ (exceeding 1 at every knob except the largest, where far-plateau leakage gives 0.977): the blind planner is not merely uninformed but *exploited* — MPC on the wall-less model plans into the phantom region, is pinned at the wall (final $x = x_{\text{wall}}$ exactly, contact rate 1.00 at every knob), and replans the same doomed plan every step, for the entire episode, scoring below random at every knob but the largest (0.00 vs 0.53). The 0.977 at $x_{\text{wall}} = 10$ is the sigmoid tail of the far plateau leaking $J_{\text{blind}} = 0.94$; the mechanism is unchanged. **The reach mechanism** appears in its cleanest form (Figure 2), with the query/trajectory distinction now visible. Gate-miss exactness is re-verified in-tests and again at gate scale in Section 5.

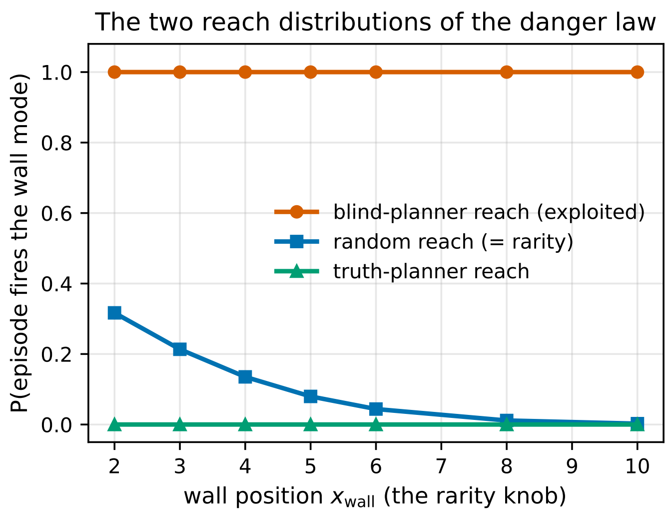


Figure 2: The two reach distributions of the danger law: the exploited planner's mode reach is flat at 1.00 across the knob; random reach (= rarity) falls two orders of magnitude; the truth planner's trajectory reach is 0.00 — the mode lives on its *query* distribution (Proposition 5), not its path.

**The residual leak is a reward artifact, not a mechanism fact — measured, not argued.** At the widest knobs the pinned planner scores *above* the uniform-random policy (cart $x_{\text{wall}} = 10$: $J_{\text{blind}} = 0.94$ against $J_{\text{rand}} = 0.53$; the three farthest pendulum stops likewise), which is why the claim above is "exploited at every knob" rather than "below random at every knob". The cause is the far plateau's sigmoid *tail*: a planner pinned at $x = 10$ still collects $1/(1 + e^{(12-10)/\text{width}})$ per step, which at width $= 0.5$ is 0.018 and over 80 steps is most of $J_{\text{blind}}$. A **sharp-plateau variant** (`scripts/continuous_sharp_plateau.py`, cart width $0.5 \to 0.2$, pendulum $0.25 \to 0.1$; sibling JSONs, the default instruments and every synthesis artifact untouched) removes the tail and settles it:

- **The exploitation claim strengthens.** On the cart the blind planner is below random at **7/7** knobs instead of 6/7, by two to thirteen orders of magnitude ($J_{\text{blind}}$ from $2.3 \times 10^{-13}$ to $2.4 \times 10^{-3}$ against $J_{\text{rand}} = 0.532$); on the pendulum, 5/6 instead of 3/6 — and 6/6 with the asymmetric variant below.

- **play_cost becomes knob-invariant to $10^{-4}$**: cart $[1.0307, 1.0309]$ (spread $1.4 \times 10^{-4}$, against the default's $5.5 \times 10^{-2}$) and pendulum $[0.9999, 1.0000]$ (spread $1.5 \times 10^{-4}$, against $6.1 \times 10^{-2}$) — a $400\times$ tightening on both, i.e. the invariance the default sweeps show approximately is exact once the tail is gone.

- **It costs no planner competence**, the pre-registered risk: a sharper plateau gives random shooting less gradient to follow, yet $J_{\text{truth}}$ moves only $17.77 \to 17.76$ (cart) and $20.08 \to 20.05$ (pendulum), and contact stays 1.00 at every knob.

- **The residue, and its fix.** Narrowing *both* plateaus starves the random policy too ($J_{\text{rand}}$ collapses to $3.6 \times 10^{-4}$ on the pendulum), which is why one knob there still reads above random: at $\theta_{\text{stop}} = 2.0$, $J_{\text{blind}} = 3.1 \times 10^{-3}$ against that collapsed baseline. The diagnosis says what to do — narrow only the *phantom* plateau, whose tail is what a pinned planner collects, and leave the real one at its default width. That variant ($\text{width}_{\text{right}} = 0.08$, per-plateau widths; `--width-right`) gives **6/6** knobs below random: $J_{\text{blind}}$ between $4.3 \times 10^{-4}$

and $7.1 \times 10^{-4}$ against a surviving $J_{\text{rand}} \in [0.0572, 0.0584]$, with $J_{\text{truth}}$ unchanged at 20.08, contact 1.00 everywhere, and play_cost in $[1.0028, 1.0029]$ — a spread of $7.1 \times 10^{-5}$, tighter still than the symmetric variant and 850× tighter than the default. So on both instruments the strong form holds: exploited, pinned, and below random at *every* knob, with play_cost invariant to $10^{-4}$, once the phantom's tail is gone.

So the mechanism does not depend on the leak, and the knob-invariance claim is the one that sharpens: it is exact, not approximate, in the instrument without a tail.

### 4.1 Robustness: the same law on a nonlinear plant

The cart's off-mode dynamics are linear, which is convenient for Section 8 but invites the worry that the phenomenology depends on it. A second instrument — a pendulum (gravity term $\sin\theta$, $\theta = 0$ hanging down) with a hard angular stop, same interface, same MPC, same two-plateau reward on $\theta$ — reproduces the identical picture with *no* re-calibration (rarity is natural here: gravity confines the random walk near the bottom, so climbing to the stop is rare):

| $\theta_{\text{stop}}$ | rarity | $J_{\text{truth}}$ | $J_{\text{blind}}$ | play_cost | blind hit | d@40 |
|---|---|---|---|---|---|---|
| 0.8 | 0.2876 | 20.08 | 0.01 | 1.002 | 1.00 | 0.000 |
| 1.0 | 0.1270 | 20.08 | 0.03 | 1.002 | 1.00 | 0.004 |
| 1.2 | 0.0509 | 20.08 | 0.05 | 1.000 | 1.00 | 0.124 |
| 1.4 | 0.0196 | 20.08 | 0.12 | 0.997 | 1.00 | 0.452 |
| 1.6 | 0.0070 | 20.08 | 0.26 | 0.990 | 1.00 | 0.748 |
| 2.0 | 0.0007 | 20.08 | 1.23 | 0.942 | 1.00 | 0.917 |

Table 3: Pendulum-with-stop: threshold law, knob-invariant exploitation (pinned at the stop in every episode at every knob), truth planner untouched by the mode. The mechanism does not care that the plant is nonlinear — only that the mode is hard and rare under the gate's measure. Rarity is the fraction of **30,000** random rollouts that fire the stop (Wilson 95% CIs for every row in the versioned JSON). The sample was raised from 3000 specifically because the widest knob was a censored zero there (0/3000); at 30,000 it resolves to 0.0007 $[0.0004, 0.0010]$, so every row is now a point estimate and every d@40 a number rather than a bound.

### 4.2 Two modes at once: a 4D bi-modal instrument

Both instruments so far are one hard boundary in a 2D state. **PatchField2D** closes both structural gaps at once — a 4D state and two independent modes — and asks what the 1D instruments cannot: does the danger law compose mode-wise? State $(x, y, v_x, v_y)$; a *scalar* action $a \in [-a_{\max}, a_{\max}]$ is mapped to a heading $\phi = \pi\,\text{clamp}(a)/a_{\max}$ so all planner machinery (which consumes scalar actions) is reused unchanged; semi-implicit Euler:

$$v'_x = v_x + (\text{gain}\cos\phi - \text{drag}\,v_x)\,dt, \quad v'_y = v_y + (\text{gain}\sin\phi - \text{drag}\,v_y)\,dt,$$

$x' = x + v'_x\,dt$, $y' = y + v'_y\,dt$. The two **modes** are circular sticky patches $P_i = \text{disc}(c_i, R)$: if $(x', y') \in P_i$ the next state is $(x, y, 0, 0)$ — the probe stops inelastically at its *previous* position with zero velocity (the 2D analogue of the wall clamp, and the structure the 2D mitigation exploits, Section 6.1). Reward is two radial sigmoid lodes, a small real one near the start and a large phantom one behind the patches; the mode-blind planner aims straight at the phantom and freezes at a patch edge. The two knobs are the patch centers' distances along and off the start-to-lode corridor ($k_1$ the near patch, $k_2$ the far one); defaults are frozen once, never tuned per cell.

Table 4 sweeps the $3 \times 3$ knob grid (`scripts/continuous_patch2d.py`; 600 rarity rollouts and 20 MPC episodes/arm per cell). The per-mode rarities separate cleanly ($r_1 \in [0.085, 0.245]$, $r_2 \in [0.0067, 0.0100]$), $J_{\text{blind}} = 0$ at every cell (the blind planner freezes at a patch edge every episode; $J_{\text{rand}} = 0.11$), and play_cost is knob-invariant at $[1.005, 1.006]$ — the same below-random exploitation as the 1D instruments, now on a 4D plant. **The danger law applies per mode and to the pair — but it does not *factor*.** Proposition 1 holds for any measurable critical event, so the per-mode gate-miss factors are $(1 - r_i)^N$ and the joint one is $(1 - r_\cup)^N$, where $r_\cup$ is the probability that a rollout contacts *either* patch (Proposition 2). We *measure* $r_\cup$ rather than compose it: writing the joint factor as the product $(1 - r_1)^N (1 - r_2)^N$ would assume the two per-mode contact events are independent within a rollout, and they are not. The measured $P(\text{both})$ ranges over $[0, 0.005]$ against $r_1 r_2 \in [0.0007, 0.0025]$ — below independence at some knobs (a rollout that freezes has spent its travel) and above it at others (reaching the far patch usually means passing the near one) — so the product misestimates the joint factor by $-17\%$ to $+12\%$ relative across the grid, in *both* directions. Both columns are in the versioned JSON (`d40_joint` and `d40_joint_indep_approx`); the table reports the measured one. Two honest notes: the $r_2$ knob is only weakly resolved at 600 rollouts (its trend is under-resolved, not flat), and the $k_1$=4 one-hit bump in $r_2$ is sampling noise rather than a knob effect.

| $k_1$ | $k_2$ | $r_1$ | $r_2$ | $r_\cup$ | $J_{\text{truth}}$ | play_cost | d@40 $P_1$ | d@40 $P_2$ | d@40 joint |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 0.2450 | 0.0100 | 0.2517 | 17.98 | 1.006 | 0.0000 | 0.6732 | 0.0000 |
| 2 | 7 | 0.2450 | 0.0083 | 0.2533 | 18.02 | 1.006 | 0.0000 | 0.7200 | 0.0000 |
| 2 | 8 | 0.2450 | 0.0067 | 0.2517 | 18.08 | 1.006 | 0.0000 | 0.7700 | 0.0000 |
| 3 | 6 | 0.1417 | 0.0083 | 0.1500 | 18.57 | 1.006 | 0.0022 | 0.7197 | 0.0015 |
| 3 | 7 | 0.1417 | 0.0083 | 0.1500 | 18.47 | 1.006 | 0.0022 | 0.7197 | 0.0015 |
| 3 | 8 | 0.1417 | 0.0067 | 0.1483 | 17.72 | 1.006 | 0.0022 | 0.7699 | 0.0016 |
| 4 | 6 | 0.0883 | 0.0083 | 0.0917 | 20.85 | 1.005 | 0.0249 | 0.7192 | 0.0215 |
| 4 | 7 | 0.0883 | 0.0100 | 0.0950 | 19.52 | 1.006 | 0.0249 | 0.6727 | 0.0185 |
| 4 | 8 | 0.0850 | 0.0083 | 0.0933 | 19.08 | 1.006 | 0.0288 | 0.7196 | 0.0200 |

Table 4: PatchField2D mechanism sweep ($3 \times 3$ knobs; 600 rarity rollouts, 20 MPC episodes/arm; $J_{\text{blind}} = 0$, $J_{\text{rand}} = 0.11$ at every cell). d@40 $P_i$ = play_cost $\cdot (1 - r_i)^{40}$ (per-mode danger); $r_\cup$ = measured probability that a rollout contacts either patch; d@40 joint = play_cost $\cdot (1 - r_\cup)^{40}$, the danger law at the joint critical event — *not* the product $((1 - r_1)(1 - r_2))^{40}$, which would assume within-rollout independence and errs by $-17\%$ to $+12\%$ here. Full table (with $J_{\text{blind}}$, $J_{\text{rand}}$, hit rates) in `docs/EXPERIMENTS.md`.

### 4.3 A second planner family: play_cost is planner-dependent

Proposition 5 predicts two branches: a blind model can change behavior only to the extent that the planner queries its disagreement region. Random-shooting MPC's constant candidates reach the distant phantom plateau in imagination and produce play_cost $\approx$1. We repeated the full 11-knob grid with a second base planner, the cross-entropy method (CEM; `scripts/continuous_cem.py`; horizon 40, 5 iterations, 64 samples, elite fraction 0.125, minimum standard deviation 0.05; one fixed setting across both instruments). The crossing columns in Table 5 are the fraction of sampled imagined trajectories that cross the omitted boundary, measured for BOTH planners with one plan from the same paired initial state per episode seed (episode-accumulated CEM fractions, which tell the same story, are recorded in the results JSON).

| instrument | knob | pc MPC | pc CEM | contact CEM | crossing CEM | crossing MPC |
|---|---|---|---|---|---|---|
| cart | 2.0 | 1.031 | 0.000 | 0.00 | 0.0309 | 0.3865 |
| cart | 4.0 | 1.031 | 0.000 | 0.00 | 0.0055 | 0.2453 |
| cart | 6.0 | 1.031 | 0.000 | 0.00 | 0.0003 | 0.1483 |
| cart | 8.0 | 1.030 | 0.000 | 0.00 | 0.0000 | 0.0773 |
| cart | 10.0 | 0.977 | 0.000 | 0.00 | 0.0000 | 0.0369 |
| pendulum | 0.8 | 1.002 | 0.009 | 0.70 | 0.1150 | 0.6392 |
| pendulum | 1.0 | 1.002 | 0.025 | 0.25 | 0.0483 | 0.5530 |
| pendulum | 1.2 | 1.000 | −0.011 | 0.00 | 0.0164 | 0.4672 |
| pendulum | 1.4 | 0.997 | −0.021 | 0.00 | 0.0053 | 0.3842 |
| pendulum | 1.6 | 0.990 | −0.021 | 0.00 | 0.0013 | 0.3039 |
| pendulum | 2.0 | 0.942 | 0.000 | 0.00 | 0.0002 | 0.2158 |

Table 5: The same certified-blind models under two base planner families (20 paired episodes/row). "pc" is blind-model play_cost within that planner family; crossing is the sampled imagined-boundary-crossing proxy for query-hit mass, one plan per episode seed from paired initial states for both planners. CEM stays near zero play cost and below MPC's crossing fraction on every row.

CEM's blind-model play cost lies in $[-0.0213, 0.0248]$ on every row — and the seed-paired 95% t-interval includes zero on all 11 rows — while its imagined crossing fraction is strictly below MPC's throughout. On the cart, truth- and blind-model CEM returns are identical and contact is zero everywhere. The nearest pendulum stops are an honest qualification: CEM contacts $\theta_{\text{stop}} = 0.8$ in 70% of episodes and 1.0 in 25%, but does not enter MPC's pinned, below-random regime; contact is zero from $\theta_{\text{stop}} \geq 1.2$. Contact is therefore not itself exploitation. This is the other measured branch of Proposition 5: the same certified-blind model is a landmine whose consequence depends on the planner's query reach, operationalizing Section 2.3's first lesson — if search does not discover the phantom, it cannot optimize toward it.

Two caveats prevent the wrong conclusion. CEM's pendulum truth return varies from 15.36 to 16.46 (versus MPC's 20.08), consistent with local optima; the comparison is blind-CEM against truth-CEM, not a claim that CEM is globally optimal. And limited reach is not knowledge or mitigation: a planner that misses a phantom distant reward can also miss a real one. This is one fixed CEM configuration, not a hyperparameter sweep.

**PatchField2D: the anticipated 2D competence gap did not materialize.** On the 4D bi-modal instrument (`scripts/continuous_cem_patch2d.py`) blind CEM is again *not* exploited: play_cost is $-0.022/+0.017/+0.020$ at knobs $(2, 6)/(3, 7)/(4, 8)$ with the seed-paired 95% $t$-interval including zero on all three rows, and CEM's imagined crossing fraction is below MPC's on every row ($0.070 < 0.208$, $0.027 < 0.149$, $0.009 < 0.094$). Here CEM is *competent in aggregate* — the 2D competence gap we anticipated did not appear — but with per-seed variance (one truth-CEM episode at $\approx 0.97$), and its blind contact rate is a uniform 0.05 (nonzero, unlike the cart's 0.00, and without positive play cost). The same low-query-reach branch of Proposition 5, one more instrument.

# 5 Axis separation: the gate fails only where the law says it can

The classic continuous-model failure axis is pervasive sub-tolerance error; the danger law's axis is a localized hard mode. A tolerance gate ($\varepsilon = 0.01$, deployment-realistic) must be shown to fail *only* on the second axis, and only at the $(1-r)^N$ rate. Five arms, one table (`scripts/continuous_axes.py`;

reveal-rarity = P(a random rollout contains a transition where truth and model differ $> \varepsilon$), 2000 rollouts; pass@40 over 300 independent $N$=40 gates; 20 MPC episodes/arm). For the hard mode, the mode-firing rarity of Tables 2–3 and the reveal-rarity here coincide as events — the mode error exceeds every deployment-realistic $\varepsilon$ (the $\varepsilon$-sweep below), so a rollout fires the mode iff it reveals a disagreement — which licenses the shared symbol $r$.[1]

| arm | reveal-rarity | $(1-r)^{40}$ | pass@40 measured | play_cost | d@40 |
|---|---|---|---|---|---|
| wall@4 omitted | 0.1351 | 0.0030 | 0.003 | 1.031 | 0.0031 |
| wall@8 omitted | 0.0103 | 0.6622 | 0.667 | 1.030 | 0.6821 |
| drag bias ×1.03 (sub-$\varepsilon$) | 0.0001 | 0.9940 | 0.997 | 0.000 | 0.0000 |
| drag bias ×2.0 (supra-$\varepsilon$) | 1.0000 | 0.0000 | 0.000 | 0.000 | 0.0000 |
| $C^\infty$ bump@4, amp 0.5 | 0.1829 | 0.0003 | 0.000 | 0.000 | 0.0000 |
| $C^\infty$ bump@4, amp 1.0 | 0.2018 | 0.0001 | 0.000 | −0.745 | −0.0001 |

Table 6: Axis separation at $\varepsilon = 0.01$, $N = 40$; d@$N$ = play_cost $\cdot (1-r)^N$, as in Table 2. Danger lives in one quadrant only: rare ∧ hard-mode. Reveal-rarity is measured on **20,000** rollouts (raised from 2000 because the sub-$\varepsilon$ arm was a censored zero there): it now resolves to 0.0001, and its predicted pass rate 0.9940 sits inside the measured $[0.9814, 0.9994]$ instead of above it.

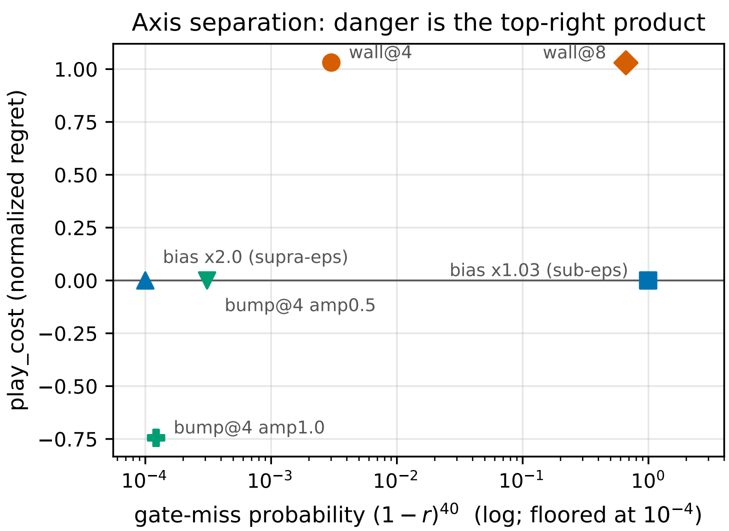


Figure 3: Table 6 as the danger quadrant: gate-miss probability (log) vs play_cost. Only the rare hard mode reaches the top right; danger is the product of the coordinates.

Readings (Table 6, rendered as the danger quadrant in Figure 3): **gate exactness at gate scale** — measured pass@40 matches $(1-r)^{40}$ in both wall rows — 0.003 vs 0.0030 and 0.667 vs 0.6622, each prediction inside the 300-gate Wilson interval. That agreement is what raising the rarity sample bought: on 2000 rollouts the wall@8 prediction was 0.6046, marginally *below* the interval's lower end (0.6115), and the gap was the rarity estimate's own noise rather than the law's. The proposition is the observed acceptance rate, not asymptotic decoration. **The gate polices the**

[1] With both samples raised, the two estimates of the same event now agree to the third decimal: mode-firing rarity 0.1352 on 30,000 rollouts (Table 2) against reveal-rarity 0.1351 on 20,000 (Table 6). At the earlier sample sizes they read 0.1430 and 0.1385 — the gap was resampling noise, as this now confirms rather than asserts.

**pervasive axis**: a global drag bias above tolerance is revealed on every rollout and never accepted; a sub-tolerance bias is accepted at 0.997 — which the law now *predicts* rather than excuses: that arm does reveal a disagreement on a thin slice of rollouts (reveal-rarity 0.0001, the extreme velocity tail), for a predicted pass rate of 0.9940 — and it is harmless at play. Verified-and-fine is a real cell, and the gate finds it. **Smoothness kills consequence, not detectability**: the $C^{\infty}$ drag bump at the wall's location has comparable rarity to the wall (0.18 vs 0.14 — it is just as *detectable*, confirming Proposition 8 is about error geometry, not about hiding from the gate, and matching the direction of Proposition 9) yet play_cost 0.000 at amplitude 0.5. At amplitude 1.0 play_cost turns *negative* ($-0.745$): the truth planner, seeing the slowdown near its horizon edge, is over-pessimistic and often settles for the small plateau, while the bump-blind planner pushes through and wins. A smooth localized omission produces planner-side timing effects of ambiguous sign; only the hard mode produces the one-way exploitation geometry (pinned, forever, below random).

**Is $\varepsilon = 0.01$ a special setting, or is the axis separation itself $\varepsilon$-invariant?** A sweep over $\varepsilon \in \{10^{-9}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 3\times10^{-2}, 0.1, 0.3\}$ (`scripts/continuous_eps_sweep.py`; Table 7 for the cart, full grid in `docs/EXPERIMENTS.md`) answers: $\varepsilon$-invariant. Mode-arm reveal-rarity is flat across the entire deployment-realistic range — on the cart, `wall@8` is bit-identically flat through the whole grid including $\varepsilon = 0.3$, and `wall@4` only dips slightly (never widens) at the top of the grid. The pervasive bias arms switch sharply at their own error scale on both instruments instead.

| $\varepsilon$ | wall@8 rarity | bias $\times$1.03 rarity | bias $\times$2.0 rarity |
|---:|---:|---:|---:|
| $10^{-6}$ | 0.0125 | 1.0000 | 1.0000 |
| $10^{-2}$ | 0.0125 | 0.0000 | 1.0000 |
| 0.1 | 0.0125 | 0.0000 | 0.0040 |
| 0.3 | 0.0125 | 0.0000 | 0.0040 |

Table 7: Reveal-rarity vs. $\varepsilon$ (cart). The mode arm is flat across the whole grid; the pervasive arms switch at their own error scale.

The pendulum replicates both halves: its mode arms are flat too (only a slight dip at the top of the grid — `stop@1.0` rarity 0.1410 at $\varepsilon \le 3\times10^{-2}$, dipping to 0.1400 at $\varepsilon = 0.1$ and 0.1240 at $\varepsilon = 0.3$), and its bias arms switch at the same error-scale boundaries as the cart's. pass@40 $\approx (1-r)^{40}$ continues to hold for the mode arms at every $\varepsilon$ in the grid, on both instruments. (The sweep's own rarity column is measured on 2000 rollouts per cell, since what it has to establish is *flatness in* $\varepsilon$ rather than a third digit; the 20,000-rollout estimate of the same quantity in Table 6 is 0.0103 against this sweep's 0.0125, one estimate's noise apart. An earlier version of this note recorded the closed-form prediction sitting marginally below the empirical Wilson lower bound; with the rarity re-measured on 20,000 rollouts that gap closes.) play_cost is not re-measured across $\varepsilon$: the model under test does not depend on the gate's tolerance, so play behavior is $\varepsilon$-independent by construction — the sweep varies only what the gate can see. The gate's $\varepsilon$ is a pervasive-error dial, not a mode-detection dial: tightening it cannot catch the hard mode, and loosening it does not widen the hole.

**The 4D bi-modal instrument replicates the flat mode arm, once per mode.** On PatchField2D the $\varepsilon$-sweep (`scripts/continuous_eps_sweep_patch2d.py`) gives *exactly* flat mode-arm reveal-rarity across the entire grid: patches-omitted 0.147, patch-1-only 0.142, patch-2-only 0.005 — each constant at every $\varepsilon$. The per-mode arms recover the per-mode rarities of the mechanism sweep (Section 4.2), so the tolerance axis is orthogonal to the mode hole on the bi-modal instrument too, and separately for each mode.

## 6 Mitigation: the exploitation is planner-mediated

The exploitation measured in Sections 4 and 5 is planner-mediated, not model-mediated, and a planner-side fix collapses it without touching the model or the gate — this does **not** contradict the danger law (the gate still certifies a wrong model; Proposition 4 is untouched). Like those sections, this one uses the hand-written instruments and blind models, not LLM synthesis. Distrust-region replanning (`src/cwm/continuous/mitigation.py`, strictly additive) compares the model's prediction against the observed transition after every real step (tol $= 10^{-6}$); a disagreement records the *position of the model's refuted prediction* — not the pre-state — as a one-sided fence, since false predictions always lie on or beyond the mode boundary. While scoring a candidate rollout, the first imagined step whose position interval crosses a fence's $\varepsilon$-band ($\varepsilon = 0.25$ cart, $\varepsilon = 0.1$ pendulum) truncates the rollout — reward kept, everything downstream dropped — which makes the fence leap-proof at any imagined speed; candidates are then ranked by (truncated return, distance to the nearest fence), which structurally prefers the real side. With zero violations this is bit-identical to plain MPC by construction (tested bitwise) — mitigation costs nothing when the model is right. Only this design is undodgeable: three earlier attempts (pre-state flee-balls; full-state point fences, dodged by the planner probing crossing velocities) were each defeated by the argmax planner acting as an adversary against the fence. The collapse is scoped to **hard-boundary hybrid modes**: the one-sided fence works precisely because a refuted prediction lies on or beyond the mode boundary, so fencing its far side cannot cut off any real trajectory. This structural fact is what these wall/stop instruments supply; less structured failure modes (soft or moving boundaries, errors not confined to a hard stop) do not obviously admit such a fence and are untested — this is a hard-boundary mitigation, not yet a general planner-side one.

| knob | pc_blind | pc_mit | first-contact step |
|---|---|---|---|
| *cart* ($x_{\text{wall}}$) | | | |
| 2 | 1.031 | 0.290 | 11.6 |
| 4 | 1.031 | 0.446 | 16.9 |
| 6 | 1.031 | 0.578 | 21.3 |
| 8 | 1.030 | 0.699 | 25.1 |
| 10 | 0.977 | 0.806 | 28.7 |
| *pendulum* ($\theta_{\text{stop}}$) | | | |
| 0.8 | 1.002 | 0.113 | 7.0 |
| 1.0 | 1.002 | 0.129 | 8.1 |
| 1.2 | 1.000 | 0.143 | 9.0 |
| 1.4 | 0.997 | 0.160 | 10.0 |
| 1.6 | 0.990 | 0.177 | 11.0 |
| 2.0 | 0.942 | 0.212 | 13.0 |

Table 8: Mitigation sweep (20 episodes/knob, both instruments). pc_blind / pc_mit = play_cost of the blind planner under plain MPC / under distrust-region mitigation. Full 11-row table with contact rates and violation counts in `docs/EXPERIMENTS.md`.

The collapse is large at every knob (Table 8): `pc_blind` stays pinned at $\approx 0.94$–$1.03$ everywhere (established in Section 4, Tables 2 and 3), while `pc_mit` never exceeds 0.81. Exactly one violation suffices to fence the mode on *every* one of the 11 rows — the mitigated planner must touch the mode once, which is identifiability operationalized: you cannot avoid what you have never seen. The residual `pc_mit` is the honest cost of that unavoidable first contact, and it grows with the lure distance, read off the first-contact step: cart $0.290 \to 0.806$ as first contact goes $11.6 \to 28.7$ of 80 steps (knob $2 \to 10$); pendulum $0.113 \to 0.212$ as first contact goes $7.0 \to 13.0$ (knob $0.8 \to 2.0$).

The cart knob = 10 row is the least favorable in the sweep (`pc_mit` 0.806, close to blind's 0.977) because the transient consumes most of the horizon — but the blind planner stays pinned *forever* there ($J$ 0.94 of $J_{\text{truth}}$ 17.77) while the mitigated planner escapes and recovers most of the horizon ($J$ 3.88): the normalized cost looks modest, the actual return does not.

**Why exactly one violation, and what sets the count in general.** The single-violation sufficiency is not a lucky measurement either — it is a covering number equal to one.

**Proposition 10** (the mitigation's violation count is bounded by a covering number)**.** *Run distrust-region replanning with band $\varepsilon$. Let $A$ be the set of mode-boundary points at which a violation can occur (the points where the mitigated planner's real step can enter the mode region). Each violation places a fence at a point of $A$, and once the accumulated fences $\varepsilon$-cover $A$, every imagined trajectory entering the mode region crosses $A$ within $\varepsilon$ of a fence and is therefore truncated — so no candidate scores the phantom and no further violation occurs. Hence the number of* coverage-adding *violations is at most $N_{\text{cov}}(A, \varepsilon)$, the $\varepsilon$-covering number of $A$.*

The two instrument families instantiate the two extremes of that bound.

- **1D: $N_{\text{cov}} = 1$, and the bound is tight.** A position or angular clamp has $A = \{x_{\text{wall}}\}$ (resp. $\{\theta_{\text{stop}}\}$) — a single point — so at most one violation is possible, and Table 8 measures a mean of exactly 1.00 on all eleven rows. The exactness of that 1.00 is the proposition, not a regularity: in one dimension the mode is separated from the reachable region by a *point*, and one fence covers a point.

- **2D: the count is the boundary's covering number, and it grows with the reachable arc.** A disc patch of radius $R$ has $A \subseteq \partial\,\text{disc}$, an arc of a circle. A fence at a boundary point covers the circle points within chord distance $\varepsilon$, an arc of angular width $4 \arcsin(\varepsilon/2R)$, so the whole circle needs $\lceil \pi/(2 \arcsin(\varepsilon/2R)) \rceil = \mathbf{7}$ fences at $R = 1$, $\varepsilon = 0.5$ (and 6 if the centers were placed optimally rather than on the boundary; both figures are brute-force verified in `scripts/circle_covering_number.py`, which also checks that one fewer does not suffice). The measured means 1.05/2.65/4.25 (Table 9) correspond to reachable arcs of 17%/43%/68% of the circle, so the farthest knob spends about two thirds of the covering budget — the "boundary-mapping transient" of Section 6.1 quantified: as the patch recedes, the planner rounds more of it before the fences close the phantom off, and it approaches the bound rather than sitting comfortably under it. (A metric covering number at *fixed* radius is the relevant quantity here because the algorithm fixes the band $\varepsilon$; it is a different question from the minimal *good* cover of a circle by contractible arcs, which is 3 with no radius constraint and is the nerve-theoretic object rather than the metric one.)

**Corollary 3** (the deployment repair cost is exponential in the boundary's dimension)**.** *Proposition 10 is dimension-free — its proof uses only that fences $\varepsilon$-cover $A$ — so instantiating it needs only a covering number. If the reachable boundary $A$ is a $p$-dimensional Lipschitz piece of diameter $D$, then $N_{\text{cov}}(A, \varepsilon) = O\big((D/\varepsilon)^p\big)$, so the number of contacts the mitigation must pay before it closes the phantom off grows exponentially in the* boundary's *dimension $p$ (not the state's). The two instruments are $p = 0$ (a point: $N = 1$, one contact, exactly as measured) and $p = 1$ (an arc: $N = O(D/\varepsilon)$, and the measured $1.05 \to 4.25$ as the reachable arc lengthens). A $p = 2$ mode surface would cost $O((D/\varepsilon)^2)$ — which is the precise sense in which the* danger *is dimension-free while the* repair *is not.*

This is also the one place where the covering-number analogue that Section 10 lists as open does appear — not on the gate side, where it remains open, but on the mitigation side, where the cost of fencing a mode is exactly a covering number of its reachable boundary. The dimensional reading is the paper's own: the danger law's $(1-r)^N$ is dimension-free, but *repairing* the model at deployment time costs $N_{\mathrm{cov}}(\partial M, \varepsilon)$, which grows with the boundary's dimension.

### 6.1 The 2D mitigation: a partial collapse that decays with mode distance

The distrust-region fence generalizes to PatchField2D (`scripts/continuous_mitigation_patch2d.py`): fences are the 2D positions of refuted predictions, which lie *inside* an unreachable patch (the stay-at-previous-position clamp forces a refuted prediction strictly inside the disc, so fencing it cannot cut off any real trajectory), and a candidate rollout truncates when an imagined step *segment* passes within $\varepsilon$ of a fence (segment-to-point distance, leap-proof); the 1D code path is preserved *behaviorally* bit-identically — the generalization rewrote the shared internals, so this is pinned by tests rather than by the diff: the no-violation case reduces to plain MPC exactly, and the 1D mitigated episodes with violations are golden-pinned to the run the sweep used (`tests/test_mitigation_1d_regression.py`). But the collapse is now **partial and decays with patch distance** (Table 9): `pc_blind` $\approx$ 1.006 falls to `pc_mit` 0.257/0.541/0.862 at knobs $(2,6)/(3,7)/(4,8)$, with mean violations 1.05/2.65/4.25 (against the 1D instruments' clean 1.0) and first contact at step 8.5/12.35/15.25. The 1D single-violation sufficiency becomes a **boundary-mapping transient**: the planner rounds one fence disc and re-contacts the patch edge elsewhere, accreting fences along the probed arc, so more of the horizon is spent mapping the curved boundary as the patch recedes. At $(4,8)$ the transient nearly defeats the mitigation (`pc_mit` 0.862 against `pc_blind` 1.006). This is the honest scope of a *hard-boundary* mitigation on a curved 2D mode — the one-sided fence still works, but a circular boundary costs a multi-contact mapping transient where a 1D clamp costs exactly one contact — not a failure of the fence.

| $(k_1,k_2)$ | pc_blind | pc_mit | mean viol. | first contact |
|---|---|---|---|---|
| $(2,6)$ | 1.006 | 0.257 | 1.05 | 8.50 |
| $(3,7)$ | 1.006 | 0.541 | 2.65 | 12.35 |
| $(4,8)$ | 1.006 | 0.862 | 4.25 | 15.25 |

Table 9: PatchField2D mitigation sweep (paired seeds). The collapse is partial and decays with patch distance: mean violations grow 1.05 $\to$ 4.25 (a boundary-mapping transient, versus the 1D instruments' single violation), and `pc_mit` approaches `pc_blind` at the farthest knob.

## 7 LLM synthesis: the danger collapses to pure identifiability

Real-LLM arms (`scripts/continuous_danger_synthesis.py`; Azure GPT-5.x mini and large[2]; $N = 40$ training rollouts which double as the gate, as in the companion paper's sweep; $\varepsilon = 10^{-9}$ pinned-integrator gate; **20 seeds/cell on the headline** $x_{\mathrm{wall}} = 8$ **cell, both sizes** (the companion paper's standard); 6 MPC play episodes/seed; per-seed JSON with the synthesized code versioned

[2]**Exact models.** "large" is the Azure OpenAI deployment `gpt-5.4` and "mini" is `gpt-5.4-mini`, both at API version `2025-04-01-preview`; the cross-family arms are `Qwen/Qwen3-Coder-30B-A3B-Instruct` (Hugging Face Inference Providers router) and Claude Sonnet (agent-relayed, Section 7). Every result JSON under `results/` records its own `model` field, so each number is attributable to a named deployment; the runs are dated 2026-07-07 to 2026-07-20 in the repository history. We write "GPT-5.x" in prose where the claim holds for both sizes.

in the repository). The contract pins the integrator (Section 2.1's equations, stated in the spec text, constants generated from the environment instance so they cannot drift); the *full* arm includes the wall clause, the *incomplete* arm omits it. Crucially, each seed logs whether the wall fired in its training sample — the identifiability event that the companion paper could not condition on post hoc.

**One design fact governs every pooled count in this paper.** The gate sample of seed index $i$ is drawn with rollout seed $10^4(i{+}1)$ *independently of which model is being run* (`scripts/continuous_danger_synthesis.py`), so at every instrument and knob the mini and large arms are synthesized from the *same* samples. Pooling the two sizes therefore doubles the number of *synthesis draws*, not the number of independent gate samples: a pooled $k/2n$ is $n$ shared samples with two model draws each. Two consequences we hold to throughout. (i) Any empirical check of the gate-miss rate $(1-r)^N$ is a per-size statement — the mode-absent/present split is a property of the sample, so it is *identical* across sizes by construction and cannot be pooled at all. (ii) For the conditional outcomes (blind-and-exploited, repaired), pooling across sizes is meaningful because the synthesis draw is independent, but a Wilson interval over the pooled count assumes independence the samples do not have; we therefore quote the per-size bound as the conservative one and mark pooled bounds as such wherever they appear.

**On the two headline cells we removed the constraint instead of caveating it.** A `--seed-offset` run shifts the sample block (rollout_seed $= 10^4(i{+}1{+}\text{offset})$), so re-running the *large* arm at offset 20 gives it a block *disjoint* from mini's. The headline cart cell and the headline pendulum knob therefore have three arms: mini on block $S_0$, large on $S_0$ (same samples, different model) and large on $S_{20}$ (different samples, same model) — two orthogonal replications instead of one. Pooling mini@$S_0$ with large@$S_{20}$ is then a pool of *independent* samples, and the Wilson interval over it is legitimate. Every bound below labelled "disjoint" is of that kind; the per-size bound is still reported for the cells where only shared blocks exist (the pendulum caught knob, PatchField2D, the ablations).

**Full arm (both sizes, 40 seeds): gate 1.000 in 0 refinement iterations, wall probes exact, play at truth parity — every seed.** The pinned-integrator premise holds with a real LLM: correct synthesis is float-exact through the sandbox, so $\varepsilon = 10^{-9}$ costs nothing and the tolerance axis is fully disarmed. As in the discrete setting, given the rule, the model translates it perfectly.

**Incomplete arm.** Three-way structure:

1. **Wall absent from the sample** (the $(1-r)^N$ event; 10/20 seeds at $x_{\text{wall}} = 8$ for each size, consistent with $(1-0.0114)^{40} \approx 0.63$): every such seed — **20/20 across mini and large** — passed the gate at 1.000, fully wall-blind on the probes, and was exploited at play: pinned at the wall, contact rate 1.0, **play_cost 0.999**. Pooling the two *disjoint* blocks — mini on $S_0$ and large on $S_{20}$ — gives **20/20** on 20 independent samples and a Wilson 95% lower bound of **0.84**; the third arm (large on $S_0$, the same 10 samples mini saw) reproduces the conditional model-for-model, 10/10. The gate-miss rate is now poolable too: 20/40 of the independent samples lacked the mode against the predicted 0.6046 (Wilson $[0.35, 0.65]$). This is the discrete headline, synthesized end-to-end in a continuous CWM: a verified, almost-everywhere-exact model that performs worse than random.

2. **Wall present, repaired:** the LLM does *not* stay blind. It reads the failing transitions and writes the true global rule — `if x2 >= 8.0:  return [8.0, 0.0]` (or the equivalent

`if x2 > 8.0:  x2 = 8.0`) — not a curve fit. At 20 seeds both sizes repaired **every** wall-present seed: **large 10/10 in 0–1 iterations** (two from the synthesis examples alone), **mini 10/10 in 0–5 iterations**. This is the divergence from the discrete setting, where rules demonstrated by example transitions were persistently not learned (translation-not-inference). A numerically-manifested discontinuity is learnable from data in a way a symbolic game rule was not.

3. **Wall present, not repaired:** at 20 seeds on the headline cell GPT-5.x produced **no stalls**, but stalls are real where they occur (the 5-seed $x_{\text{wall}} = 4$ cell; the Qwen cross-family arm below). They are not near-misses of the rule but **superstitious local patches**: e.g. `if abs(x2 - 4.0) <= 0.15 and abs(v2) <= 2.5:  x2 = 4.0; v2 = 0.0` (verbatim from the $x_{\text{wall}} = 4$ cell, mini seed 20000, gate 0.974), or Qwen's `if x2 >= 8.0 and v2 <= 0.0:  ...` — clamps fitted to the *observed manifestation* of the mode (low-speed, near-wall contacts), which mispredict other approaches to the wall. **The gate rejected every one of them** (gate 0.49–0.999, never 1.000).

**Cross-family spot-checks (two families).** *Qwen* (HF router, `Qwen/Qwen3-Coder-30B-A3B-Instruct`, 3 seeds): the full arm is clean (3/3 gate 1.000, blind 0.0), so the pinned-integrator premise is not GPT-specific; on the incomplete arm the identifiability branch reproduces (the 1/3 wall-absent seed is gate-1.000 wall-blind and exploited, play_cost 0.999), but Qwen *repaired neither* of its two wall-present seeds (gate 0.999 and 0.491, both superstitious patches the gate refused). *Claude* (Sonnet, agent-relayed via the companion paper's protocol on `scripts/continuous_claude_step.py` — verbatim pipeline messages relayed to fresh, context-free instances per message, same $\varepsilon = 10^{-9}$ gate and MPC play as the API arms; seeds 10000/20000/30000 plus one full control per instrument; `results/continuous_claude_relay.json`): both full controls are clean (2/2 gate 1.000, blind 0.0, play_cost 0.0), and both mode-absent seeds are certified fully blind and exploited (cart play_cost 0.999, pendulum 0.995) — the identifiability event fires family-independently, as Proposition 4 requires. On repair Claude is neither GPT-5.x nor Qwen: cart seed 20000 repaired the exact one-sided rule in 1 iteration, and cart seed 30000 repaired at iteration 5 after a period-2 oscillation (symmetric $\pm 8$ walls $\to$ both removed $\to$ symmetric again $\to$ removed $\to$ one-sided correct), but pendulum seed 20000 was certified in 1 iteration carrying an invented mode (next paragraph) and pendulum seed 30000 stalled at 5 iterations (gate 0.9972), oscillating between the symmetric-stop and no-stop artifacts without ever finding the one-sided rule. So across three families the mode-absent blind-and-exploited event fired for every one (it is a property of the sample; Proposition 4), while repair-from-data is model-dependent *in mechanism, not merely in rate*: GPT-5.x repairs every revealed mode exactly (62/62 pendulum, 20/20 cart), Qwen repairs none (superstitious local patches), and Claude repairs most but through a *symmetry prior* — it generalizes one-sided boundary evidence into a symmetric pair of boundaries.

**A fourth artifact class: certified with an invented mode unfalsifiable by its gate sample.** Claude's symmetry prior produces an outcome beyond the correct/blind/superstitious-patch trichotomy. Where the training sample covers the invented side, the gate refutes the phantom and the memoryless refine loop oscillates (cart seed 30000's period-2 cycle above; pendulum seed 30000's stall at gate 0.9972). But where the sample is *silent* on the invented side, the phantom is unfalsifiable: pendulum seed 20000 was certified at gate 1.000 in a single iteration while carrying a phantom symmetric stop at $\theta = -1.4$ that this seed's rollouts never reach (verbatim `elif th2 < -th_max:  th2 = -th_max; om2 = 0.0`, the second branch of a symmetric pair, with `th_max =`

`1.4` its own constant) — a verified artifact, exact on every sampled transition, that nonetheless encodes a hard stop which does not exist. This is a clean natural experiment: the *same* symmetric artifact was refuted at pendulum seed 30000 and certified at seed 20000, the only difference being whether the sample covered $\theta < -1.4$. It is Proposition 4's prior caveat measured directly — on inputs the sample never touches, the artifact's content comes from the model's prior, and the gate cannot police it. Note the classification did not flag it: the `mode_blindness` probe scored 0.0 (correct) because it probes only the true $+1.4$ mode's region; code inspection, not the probe, caught the invented mode (Section 10).

**Honesty notes (Claude arm).** (i) Agent-relayed means an agent scaffold over a subscription transport, not an API, though the relayed messages are byte-identical to the pipeline's; in the two multi-iteration cells a handful of refinement replies prefixed a one-line explanation before the code block (two distinct sentences, repeated across the oscillation) despite the output-only-code instruction — recorded, and the code block still parsed; no relay was refused this time (contrast the discrete probe). (ii) The refine loop is memoryless in the API arms too — `refine_continuous` sends a single user message per iteration (`src/cwm/continuous/contract.py`) — so the oscillation is protocol behavior, not a relay artifact. (iii) $n$ is small: 3 seeds plus one control per instrument, one alternate family.

The branches compose into the paper's central claim. With the mode in the data, the synthesize–refine–gate loop behaved soundly *on every transition the sample covered*: it recovers the exact mode or refuses the artifact (sample-covered exactness is the acceptance criterion itself; the empirical, code-inspected regularity — not a theorem — is that every accepted artifact's off-sample content was also the true mode, except Claude's phantom pendulum stop, which errs only where its sample is silent: the Proposition 4 prior caveat, not a gate failure). With the mode absent, no loop can help (Proposition 4: the sample carries no evidence for the mode — though a prior or the specification still could supply it), and the acceptance of a blind artifact is not a failure of the LLM, the refinement, or the gate implementation — it is the sampling event whose probability is exactly $(1-r)^N$. The discrete danger law had a provable core plus an empirical residual (the rule not learned even when shown); on the continuous 1D instruments the residual *can* vanish — it does for GPT-5.x, which repairs every revealed clamp — so **the law becomes the entire failure surface** for a capable-enough synthesizer, and shrinks toward it even for a weaker one (this is geometry-scoped: Section 7.1 shows the residual returns on a 2D circular mode, where even GPT-5.x does not repair). The actionable consequence sharpens correspondingly: in this regime, spec completeness and gate-sample coverage are not two independent worries — coverage *is* the dominant worry, because the synthesis loop repairs what the sample reveals.

(Scope: 20 seeds/cell on the headline cell, both GPT-5.x sizes; the wall-absent conditional is 20/20 across sizes and consistent across three independent runs — the 5-seed first run, the 20-seed tightened run, and the 3-seed Qwen run; the wall-present repair rate is 20/20 for GPT-5.x on this cell, 0/2 for Qwen, and mechanism-dependent for Claude (exact on both cart seeds, one via a period-2 oscillation) — model-dependent, small-$n$ on the cross-family arms. LLM synthesis is stochastic across calls; the three-way structure and the identifiability conditional are stable run-to-run, per-seed iteration counts are not.)

**Second-instrument robustness (pendulum-with-stop, Section 4.1, 20 seeds/cell, both sizes).** The synthesis arm is not cart-only. Running the identical pipeline on the nonlinear pendulum — headline $\theta_{\text{stop}} = 1.4$ (rarity 0.019) and caught $\theta_{\text{stop}} = 1.0$ (rarity 0.128; "caught" labels a cell whose mode is common enough that the gate sample essentially always contains it, so the identifiability event essentially never fires) — reproduces every branch, including a 3-seed Qwen cross-family spot-check at the headline knob (Table 10).

| cell (20 seeds each) | full | absent → blind & exploited | present → repaired (stalled) |
|---|---|---|---|
| mini $\theta_{\text{stop}} = 1.4$ | 20/20 | 9 → 9 (pc 0.995) | 11 → 11 (0) |
| large $\theta_{\text{stop}} = 1.4$ | 20/20 | 9 → 9 (pc 0.995) | 11 → 11 (0) |
| mini $\theta_{\text{stop}} = 1.0$ | 20/20 | 0 → — | 20 → 20 (0) |
| large $\theta_{\text{stop}} = 1.0$ | 20/20 | 0 → — | 20 → 20 (0) |
| Qwen $\theta_{\text{stop}} = 1.4$ (3 seeds) | 3/3 | 1 → 1 (pc 0.995) | 2 → 0 (2 stalled @0.9997) |

Table 10: Pendulum synthesis cells (20 seeds/cell, both GPT-5.x sizes; 3-seed Qwen spot-check at $\theta_{\text{stop}} = 1.4$). $k \to m$ = of the $k$ seeds in that branch, $m$ had the stated outcome (blind & exploited, resp. repaired); the parenthetical count is stalled seeds; pc = play_cost.

Pooled across both knobs and both sizes (Table 10), every mode-absent occurrence was blind and exploited at play_cost 0.995 (headline knob, disjoint blocks: 15/15 on 15 independent samples, Wilson 95% lower bound 0.796; the same-sample pair mini/large@$S_0$ adds 9/9 each) — the same fixed-point exploitation as the cart's play_cost $\approx 1$ — and GPT-5.x repaired **62/62** mode-present seeds to the exact angular clamp (verbatim at the headline knob: `if th2 >= 1.4:  return [1.4, 0.0]`), 0 stalls (Wilson 95% lower bound 0.942, pooled over two knobs and two sizes on 31 shared samples; the per-size, per-knob cells are 11/11 and 20/20). Qwen reproduces the mode-absent blind-exploited event but stalls on both its mode-present seeds, the same superstitious-patch signature as on the cart; Claude's agent-relayed spot-check on this instrument produced the phantom-mode artifact discussed in Section 7 (a certified stop at $\theta = -1.4$ its sample never reaches, plus a stall at the other mode-present seed). Repair is model-dependent; identifiability, being a property of the sample, is not — on this instrument too. The mechanism was already validated on two instruments (Section 4.1); the synthesis result now is as well: a nonlinear plant with an angular, not positional, hard stop reproduces the same danger law and the same repair capability, so the repair finding is not a cart artifact.

### 7.1 The repair finding inverts on a 2D circular mode

Everything above — and the (b)-residual-vanishes claim of the Introduction — was measured on *one-dimensional* hard boundaries: a position clamp (cart) and an angular clamp (pendulum), on which GPT-5.x repaired every revealed mode (**106/106**, over 65 distinct gate samples: the shared-block draws plus the two disjoint-block arms of Section 7). PatchField2D asks whether that repair survives a genuinely 2D mode — a circular boundary $(x' - c_x)^2 + (y' - c_y)^2 \leq R^2$ — and whether an artifact can repair one patch while staying blind to the other (**partial repair**, which the design predicted and per-mode blindness makes measurable). We ran the identical pipeline (Azure GPT-5.x mini and large, 20 seeds/cell, $N = 40$, $\varepsilon = 10^{-9}$) on two bi-knob cells, $k = (3, 7)$ and $k = (5, 9)$, classifying each seed by which modes its training sample contained {miss both, see one miss the other, see both}. The full arm is clean (20/20 per cell, both sizes).

**We predicted partial repair; we measured none.** Across the **66** see-one-miss-the-other seeds (64 see-$P_1$-miss-$P_2$ plus 2 miss-$P_1$-see-$P_2$), **zero** produced a partial-repair certificate — not one artifact repaired the seen patch and stayed blind to the unseen one while passing the gate. More strongly, of the **76** incomplete-arm seeds whose sample contained *at least one* mode, **0/76 recovered the circular disc rule at all**.

**The mechanism: translation succeeds, induction of the boundary collapses.** A 76-artifact code inspection locates the failure precisely. *Plant translation succeeds*: ≈74/76 artifacts reproduce the exact 4D semi-implicit integrator and the reward. What collapses is *induction of the 2D circular boundary.* The dominant modal failure (38/76) is **dimensional reduction**: the disc is modeled as a *half-plane* — a 1D CartWall-style clamp at the right location but the wrong shape, e.g. substituting `if x2 > 4.0: ...` for a circle — learning the freeze mechanic but not its circular trigger. The remaining classes: 20/76 pure-blind (no patch logic), 9/76 superstitious local patches, and 9/76 disc-form attempts, none of them correct. The $\varepsilon$-exactness alternative — that correct-form discs merely failed the $10^{-9}$ arithmetic gate — is **falsified**: no correct-form disc failed on arithmetic; the discs that failed were the wrong *shape.* A *behavioral* audit confirms this hand inspection on every claim (`scripts/patch2d_artifact_audit.py`: probe each artifact's `step()` on a state grid, classify the shape of its deviation-from-integrator set): 39/76 half-plane-form, 13 behaviorally blind, 8 disc-form, 7 measure-zero textual traps, 4 point, 3 square-form, 2 bounded-other; integrator arithmetic exact in 74/76; and — the point that matters for partial repair — **no artifact *encodes* its seen patch**. On the see-one-miss-the-other branch (the 66 seeds where a partial repair is even definable), 28 freeze sets do *contain* the seen patch (coverage $> 0.9$), but they contain it the way a half-plane contains a disc: their frozen area runs from 15× to 81× the patch's own ≈1.2% share of the probed box (median 61×). Exactly one artifact is even patch-*selective* — seed 180000 covers its seen patch and leaves the unseen one below 0.1 — and it does so with a *half-plane* freezing 31× the patch area, i.e. by where its threshold happens to fall between the two patches, not by encoding a region; its gate score is below 1 like every other see-one artifact. The missing partial repairs were therefore never attempted as *regions* at all, rather than attempted and refused by the gate. The two class counts differ by exactly one artifact (39 vs 38 half-plane, 8 vs 9 disc-form) and the difference is a *syntax-versus-behavior* reading, not a disagreement about any artifact's correctness: two artifacts (mini $k = (3, 7)$, seeds 130000 and 150000) write a *radial* predicate anchored at the reward lodes rather than at a patch — `if hypot(x-(-6),y) <= 2 or hypot(x-12,y) <= 2` in one, a reward-threshold freeze in the other — so hand inspection reads the disc in the source while the audit reads the deviation set, which is unbounded inside the probed box. Both readings agree that neither artifact encodes a patch; only the label moves.

**Ablation 1: richer prompting and 3× budget do not restore repair — they change the failure class.** The natural confound is that the prompt was too poor and the budget too small. The strongest joint treatment (120 examples, 40 failure lines, describe-the-region-first guidance with an explicit "the region need not be a 1D threshold" de-bias, and 15 refine iterations; identical samples to the original run) yields **0/40 repair** at $k = (3, 7)$, both sizes. The guidance *works as prompt engineering*: the half-plane reduction disappears (0/40, versus 21/40 in the matching base cells) and the artifacts now write bounded 2D regions — rotated ellipses (∼15/40), rectangles (∼10/40), unions of micro-discs (∼5/40) — but none is the true disc. The new failure class is *evidence-hull fitting* with two compounding errors: they fit the hull of the *observed* freeze positions (the pre-freeze crescent hugging the disc's reachable west boundary from outside: fitted centers pulled west to $x \approx 2.3$ against the true $x = 3$, i.e. dragged onto the evidence at the patch's west edge), and 36/40 condition on the current position rather than the landing $(x_2, y_2)$ — the causal variable of the rule. The sample itself then refuses every wrong shape.

**Ablation 2: a zero-curvature square falsifies the curvature reading.** If the failure were about *curved* boundaries, an axis-aligned square patch (a Chebyshev ball, $\max(|x-c_x|, |y-c_y|) \leq R$ — a `max`/`abs` predicate, no quadratic) should be repairable. It is not: with the same pipeline at

$k = (3, 7)$ the full arm is clean (20/20 at zero refine iterations, both sizes — translating the square clause is as easy as the disc's) and the incomplete arm gives **0/40 repair** (every sample mode-containing; best gate 0.9962). The classes mirror the disc's, *reflected*: the dominant failure is again dimensional reduction (the square's west edge as a 1D threshold, `if x2 >= 2.0`), and several artifacts commit the *inverse* error — writing *discs* on square evidence (radial `hypot` guards at the edge midpoint, micro-disc unions), plus reward-anchored superstitions (freeze zones invented at the lodes). Zero artifacts write the true box or its `max/abs` form. Pooled across the three campaigns: **0/156** mode-containing seeds recover the 2D region — which, in the shared-sample accounting of Section 7, is 78 distinct gate samples each given two independent synthesis draws (38 disc + 20 square + 20 guided). For a *negative* result the pooling is the mild direction: each draw is an independent opportunity to repair, and none took it.

**Ablation 3: a second family fails in the same classes.** The three campaigns above are GPT-5.x only, so the collapse could be one family's idiosyncrasy. A Claude arm on the headline cell says otherwise (agent-relayed as in Section 7; Sonnet, $k = (3, 7)$, three mode-containing seeds — two see-$P_1$-miss-$P_2$, one seeing both — plus one full control, same $N = 40$ sample, same $\varepsilon = 10^{-9}$ gate, same five-iteration memoryless refine loop). The full control is clean at iteration 0 (gate 1.000, both discs written on the *landing* position, blindness 0.0, play_cost 0.0): translation is not the problem here either. No incomplete seed repaired (0/3), and the *trajectory* carries more information than the outcome: each seed spends its five iterations in a **period-2 cycle** between the pure-blind artifact and a wrong template, returning to its iteration-0 gate value exactly on every even iteration (seed 10000: 0.9934 blind → 0.9591 disc-on-*current*-position → 0.9934 → 0.9284 half-plane → 0.9934 → 0.7044 reward-threshold; seed 20000: blind 0.9962 ↔ half-plane 0.9406, three times over; seed 30000: 0.9966 → 0.6750 reward-zone → 0.9966 → 0.8406 half-plane → 0.9966 → 0.5328 $y$-band). The template set is the one the GPT-5.x campaigns exhibited — 1D threshold, radial disc on the *wrong* (current, not landing) variable, reward-landmark zone, axis-aligned band — and **no incomplete iteration in any seed wrote a disc on the landing position**, the form this same model writes immediately when the contract states it. The *constants* replicate Ablation 1 quantitatively: every half-plane sits at $x = 2.0$, the disc's west edge — "the right location, the wrong shape" — and the one radial attempt fits center $(2.328, -0.135)$ with radius 0.824, i.e. the hull of the observed freeze crescent: the true patch is the disc of radius 1 about $(3, 0)$, spanning $x \in [2, 4]$, while the fitted one spans $x \in [1.50, 3.15]$ — *dragged west onto the evidence*, so it covers the patch's west half, misses $x \in [3.15, 4]$ entirely, and freezes free space over $x \in [1.50, 2]$. Its center sits inside the true patch but at the wrong place (0.69 from the true center), with radius 0.824 against 1. That is the same $x \approx 2.3$ signature the guided GPT-5.x artifacts produced. Per-iteration gate accuracies and rule classes are re-derived from the versioned transcripts by `scripts/claude_relay_ledger.py`. Two honest notes. (i) $n = 3$ seeds, one alternate family on the *incomplete* arm: this closes the "GPT-only" confound, it does not sweep models. A third family (Qwen) was launched on this cell and **aborted after its three full-arm cells** when that provider's inference credits ran out, so it contributes a translation control and nothing else — but that control is clean and worth recording: 3/3 gate 1.000 at zero refinement iterations, both discs written exactly on the landing position, per-patch blindness 0.0. Across all three families, then, *stating* the region rule is easy and *inducing* it from data is what fails; the induction evidence on this instrument is GPT-5.x's 156 seeds plus Claude's 3. (ii) The relay caveats of Section 7 apply, plus one specific to this arm: the relayed instances were instructed not to use tools or read files, so each artifact is a function of the pipeline message alone (the ground truth lives in this repository, so an instance reading it could have copied the true centers and radius instead of inducing them). The wrapper text is recorded verbatim in `results/claude_relay_transcripts/patch2d_k3_7_RELAY_FRAMING.txt`, together with the post-hoc leak check: no incomplete artifact wrote the true disc form at all, and none names the unseen far patch — the signature repository access would have left.

**The gate is all-or-nothing, and stays sound.** Certification occurred *iff* the sample missed *all* modes: the only certified incomplete artifacts were the **4/80** miss-both seeds (2 per size at $k = (5, 9)$), blind on both patches and exploited at play_cost 1.095, contact rate 1.0 — the doubly-blind fixed point, at the joint-miss event of probability $(1 - r_\cup)^N$ (Section 4.2: measured for the pair, not factored). Every see-one-miss-the-other artifact was *rejected*, not certified: the all-or-nothing gate refused every partial artifact rather than certifying a half-repair. Certification therefore stayed sound in the strict sense — no wrong artifact passed on a transition its sample covered. Two cautions on reading the failing gates: a high near-miss score (e.g. 0.9997) reflects the *rarity of a mode contact* in a short rollout, *not* a near-repair, so more refinement iterations would not populate a partial-repair branch that does not exist; and the mini/large partitions are seed-identical by construction (shared seeds — the design fact of Section 7, which holds at every instrument, not only here), so the two sizes are not independent replications.

**Repair-from-data is geometry-dependent — and the axis is a template prior, not curvature.** The arc is the paper's own: on 1D clamps the discrete paper's (b)-residual *vanished* (GPT-5.x repaired 106/106); on the 2D modes it *returns* (0/156). Repairing an omitted hybrid mode from data is therefore **geometry-dependent**, and the two ablations locate the axis: it is *not* boundary curvature (the flat square fails identically, with the errors reflected) and *not* prompting or budget (the strongest treatment fails, differently). The mechanism the three campaigns agree on is a **template prior**: the synthesizer selects from a small library of low-descriptive-complexity region forms — 1D threshold, radial ball, reward-landmark zone, evidence hull — and keeps whatever locally fits, under-weighting the evidence's actual geometry in *both* directions (flattening curves, curving flats). Dimensional reduction is one face of this prior; the square run exposes the other. Induction of a 2D generative boundary from one-sided contact evidence is exactly what the prior replaces, and the danger law's empirical exhaustiveness — "the residual vanishes, so coverage is the whole game" (Section 7) — is correspondingly **geometry-scoped**: it holds for the 1D clamps and fails on 2D regions, where a capable synthesizer translates the plant exactly yet substitutes a template for the mode. What does *not* change is the provable core: identifiability and the gate-miss law hold verbatim, certification stayed sound, and the danger still lives exactly in the joint-miss event. The residue is that on 2D geometry the loop no longer repairs what the sample reveals — so the discrete paper's exhaustiveness claim, sharpened to a boast on the 1D instruments, is scoped back by the very move (harder geometry) that was supposed to test it.

## 8 Smooth learners cannot localize: both halves, measured

If localization is representational, two things must be checkable on non-code learners trained on the *same data* (`scripts/continuous_smooth_probe.py`; the two most favorable smooth learners: closed-form linear least squares — off the wall, the dynamics are *exactly linear*, so this is the smooth best case — and a small tanh MLP, probe-grade):

**Identifiability is learner-independent, live.** On the wall-free sample the linear model recovers the off-mode dynamics to $10^{-15}$, **passes the $\varepsilon = 10^{-9}$ gate**, and is exactly as wall-blind as the synthesized blind code (probe error 4.18: it predicts straight through the wall). Proposition 4

| model | trained on | off-mode err mean / max | probe err | gate $10^{-9}$ | gate $10^{-2}$ |
|---|---|---|---|---|---|
| linear-LSQ | wall-free | 3.6e−15 / 1.7e−14 | 4.18 | **PASS** | **PASS** |
| linear-LSQ | wall-data | 1.9e−03 / 1.2e−02 | 4.17 | fail | fail |
| MLP h=8 | wall-free | 3.5e−03 / 5.0e−02 | 4.20 | fail | fail |
| MLP h=8 | wall-data | 6.0e−03 / 4.9e−02 | 4.19 | fail | fail |

Table 11: Smooth learners on the synthesis samples ($x_{\text{wall}} = 8$). “probe err” is the wall-region probe error (4.2 ≈ predicting straight through the wall).

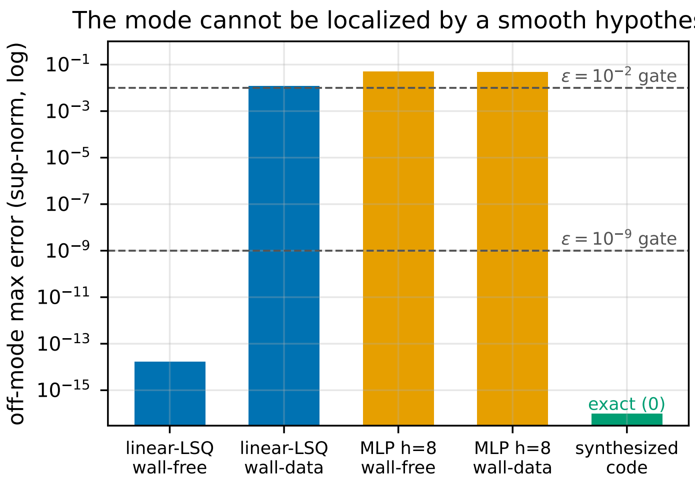


Figure 4: Off-mode max error (log) by model and training data. The wall-free linear fit sits at float noise and passes both gates (blind); four contact rows tilt it twelve orders of magnitude; synthesized code is exactly zero off-mode.

instantiated on a second hypothesis class: in the gate-miss event, *any* accepted learner is blind. The $(1-r)^N$ hole is not an LLM property.

**With the mode in the data, code and smooth part ways.** Four contact rows out of 3200 tilt the linear fit by **twelve orders of magnitude** off-mode (1.7e−14 → 1.2e−02 max; Figure 4) — it fails both gates *and still* has the mode wrong (probe 4.17). The smooth hypothesis cannot put the error on the mode; it leaks everywhere (Proposition 8’s geometry, observed as a least-squares tradeoff). The synthesized code, on the *same sample*, wrote the exact clamp and passed at float precision. The MLP combines both axes: a pervasive ∼5e−3 floor that no gate accepts, and a never-learned mode.

Consequence for the CWM paradigm: the paradigm *creates* in continuous control the localized failure class that the learned-model literature says does not dominate there — and, by the same representational fact, also creates the exact-repair capability that learned models lack. Code cuts both ways, and the gate’s sampling coverage decides which way.

# 9 Related work

*Objective mismatch and model exploitation in model-based RL* (Lambert et al., 2020; Janner et al., 2019; Hafner et al., 2020): prediction accuracy and control performance diverge for learned continuous models; planners exploit model errors. Our contribution to that conversation is the *verified-*

*but-wrong localized* regime, which smooth learned models cannot even represent (Proposition 8, Table 11), plus a closed-form acceptance-failure law confirmed at gate scale. *Hybrid systems:* mode detection and identification of piecewise/hybrid dynamics is classical (Paoletti et al., 2007; Bemporad and Morari, 1999); our question is not identifying the mode but what a *sampling verifier certifies* when the mode is missed, and what a planner then does. *Code World Models* (Lehrach et al., 2026) and code-as-policy (Liang et al., 2023; Gao et al., 2023) supply the paradigm; the companion paper (Aguilar Martín, 2026) is the discrete study this paper extends — transferring its provable core, showing its empirical residual (translation-not-inference) does not transfer, and adding the representational theorem separating code from smooth hypotheses. *Property-based testing and rare events* (Claessen and Hughes, 2000; Rubinstein and Kroese, 2017): the gate is random testing of a program against an oracle; $(1-r)^N$ is the standard rare-input coverage gap, here with the planner as the adversary that finds it. *Falsification and rare-event testing of hybrid and cyber-physical systems* (Annpureddy et al., 2011; Corso et al., 2021): this literature searches for inputs that drive a hybrid system to violate a specification; our gate is the passive dual — uniform random testing whose blind spot is precisely the rare mode a falsifier would hunt for, and the $(1-r)^N$ factor quantifies the coverage gap that random testing leaves. *Runtime monitoring and shielding for safe RL and control* (Alshiekh et al., 2018): a synthesized shield overrides unsafe actions using an a-priori specification; our distrust-region fence is instead deployment-time feedback, built from observed prediction failures rather than a given safety automaton, and it corrects a certified-but-wrong model rather than an unsafe policy. *Robust MPC and planning under model uncertainty* (Rawlings et al., 2017): robust and tube-based MPC guarantees performance against *bounded* model error; the failure here is orthogonal — the error is unbounded but localized to an omitted mode and is certified away by the gate, so robustness margins tuned to the off-mode error do not see it. *System identification of contact-rich and discontinuous dynamics* (Paoletti et al., 2007; Fazeli et al., 2017): estimating parameters and contact forces of rigid bodies undergoing frictional contact is a mature sysid problem; our repair-from-data result is the code analogue — given contact transitions in the sample, the synthesizer recovers the exact switching rule, whereas a smooth fit cannot localize it.

## 10 Limitations and Honest Assessment

**Three instruments; repair is geometry-scoped, mitigation decays with mode distance.** Cart-with-wall, pendulum-with-stop, and PatchField2D now span 1D and 4D state and one and two modes, so the earlier "one dimension, single stationary boundary" limitation no longer holds. The *mechanism* survived every extension — it is measure-theoretic — and so did strict gate soundness (the all-or-nothing 2D gate certified no partial artifact). The honest residue is now sharper and different. **Repair-from-data collapses on 2D region boundaries, curved or flat**: GPT-5.x repaired every 1D clamp (106/106) but recovered the 2D region rule in 0/156 mode-containing seeds pooled over the disc, the square ablation, and the guided/3×-budget treatment (Section 7.1). The prompting/budget and curvature outs are now *measured negatives*; what could restore repair — two-sided (inside) evidence, active boundary probing, or richer evidence summaries — is the subject of ongoing follow-up work. The **2D mitigation** likewise generalizes but only partially, its collapse decaying with mode distance as a boundary-mapping transient that nearly defeats the fence at the farthest knob (Section 6.1). Contact-rich manipulation, moving boundaries, and 3+ modes remain future work.

**Two planner families, one fixed configuration each.** Random-shooting MPC occupies the high-query-reach/high-play-cost branch of Proposition 5; CEM occupies the low-query-reach/near-zero-play-cost branch (Section 4.3). This is stronger than a one-family result but not planner-universal: CEM was measured at one prototype-fixed setting, with no hyperparameter sweep, and its pendulum truth returns expose local optima. Its non-exploitation is reach-limited, not knowledgeable or safe — the same search that misses phantom reward can miss real distant reward. Gradient-based shooting and tree search (Coulom, 2006; Kocsis and Szepesvári, 2006) remain untested. Distrust-region replanning (Section 6) is a mitigation layered on MPC, not a third family; its hard-boundary guarantee and its behavior with other planners remain untested.

**Synthesis cells are modest, and the two sizes share their samples.** The gate sample is drawn from the seed index alone, so mini and large are synthesized from identical samples at every instrument and knob (Section 7): every pooled "both sizes" count is $n$ distinct gate samples with two synthesis draws each, and the distinct-sample $n$ is half the pooled one — 10 mode-absent samples on the cart headline cell, 31 mode-present across the two pendulum knobs, 38 mode-containing on the PatchField2D disc cells, 20 on each ablation campaign. Per-size Wilson bounds are quoted for exactly this reason wherever only shared blocks exist. On the two *headline* cells the constraint was removed instead: the large arm was re-run on a disjoint seed block (`--seed-offset 20`), so there the pooled counts are over independent samples, the Wilson bounds are legitimate (0.84 cart, 0.796 pendulum), and even the $(1-r)^N$ gate-miss check becomes poolable (20/40 independent cart samples lacked the mode against a predicted 0.63). With that accounting: 20 seeds/cell on the headline $x_{\text{wall}} = 8$ cart cell for both GPT-5.x sizes, plus a 3-seed Qwen cross-family spot-check; the pendulum arm adds two knobs (headline and caught) at the same 20 seeds/cell, both sizes, plus its own 3-seed Qwen spot-check. The wall/mode-absent conditional is 20/20 across cart sizes and three runs and 18/18 across pendulum sizes: per size that is 10/10 (Wilson lower bound 0.72) and 9/9 (0.70), the bounds we rely on, against pooled 0.84 and 0.824 whose trials share samples. The GPT-5.x repair rate is 20/20 on the cart headline cell and 62/62 pooled on the pendulum (31 distinct samples). The cross-family arms are small-$n$ (3 seeds plus one control per instrument per family) and cover two alternate families (Qwen, and Claude agent-relayed), not a sweep of models — and their coverage is uneven across instruments: both families were run on the two 1D instruments, but on PatchField2D only Claude has an incomplete arm — the Qwen run there aborted after its three full-arm cells when that provider's inference credits were exhausted, so it is a translation control (3/3 exact) with no induction data, a gap of circumstance rather than of judgement; they show repair is model-dependent *in mechanism* — Qwen 0/2 mode-present (superstitious patches), Claude repairing most but via a symmetry prior that oscillated on two seeds and certified one phantom, unfalsifiable mode. The PatchField2D synthesis arm adds two bi-knob cells (20 seeds each, both GPT-5.x sizes) plus two ablation campaigns (region-guidance 3× budget; zero-curvature square — 20 seeds each, both sizes); its finding is a *negative* one — 0/76 repair of the circular mode, 0/66 partial repair, 0/156 pooled, plus 0/3 in a Claude cross-family arm on the same cell whose failure classes match (Section 7.1) — so repair-from-data is not only model-dependent but geometry-dependent, via a template prior.

**The mode-blindness probe covers only the true mode's region.** By construction `mode_blindness` (`src/cwm/continuous/contract.py`) fires probes where the truth's mode is active, so it certifies whether the *revealed* mode is encoded but is blind to a mode the model *invents* elsewhere: Claude's phantom pendulum stop (Section 7) scored blindness 0.0 while carrying an extra, non-existent stop at $\theta = -1.4$. Detecting invented modes needs code inspection, or probes seeded outside the sampled region — the classification alone does not suffice, and our phantom-mode finding rests on reading the certified code, as the companion paper's artifact-level analysis does.

**The MLP is a probe, not a baseline.** It substantiates the representational point at $h$=8/pure-Python scale; a tuned modern dynamics model would have a lower floor but the same structural inability to be bit-exact off a mode at $\varepsilon = 10^{-9}$ (an argument, not yet a measurement, at scale).

**Proposition 8 bounds geometry, not probability.** The ball is metric; converting to gate-visitation probability needs the gate measure on that ball, which is instrument-specific. The empirical complement (Table 11's twelve-orders tilt) carries the quantitative weight here.

**Coverage certificates transfer, but only to the smooth case.** The companion paper's guarantees enumerate information sets, and that enumeration has no continuous analogue. What does transfer is the covering-number version: Proposition 7 certifies $\sup_U \|f - \hat{f}\| \leq \varepsilon + 2L\rho$ for $L$-Lipschitz pairs once the gate $\rho$-covers $U$, with the sample size a covering number over a visitation density, and on the deployed cart gate it excludes any Lipschitz model carrying the wall's own error magnitude (4.2 against bounds of 1.57 rigorously, 0.43 optimistically). The honest residue is that this is exactly the case the paper is *not* about: the hybrid mode has unbounded local Lipschitz constant, so no finite $L$ makes the certificate apply to it, and the danger the paper measures lives entirely in that gap. One gap remains and one is closed. Closed: the within-rollout dependence, which does not need a mixing argument — the action's independence from the state makes the per-rollout survival probability exact and the i.i.d.-ness of rollouts makes it factorize, and doing so buys 2% ($1.57 \to 1.53$) rather than the factor of four an all-steps-independent reading suggests, because the binding constraint is the number of rollouts reaching the worst cell. Also closed: the density is now derived at *every* step $t \geq 2$ (the last two actions make the step-$t$ law absolutely continuous with an exact constant $1/0.036$), which extends the certificate to a region 5.7× larger while still excluding the hard mode — and the coverage threshold it predicts is measured to be tight to one grid step. Both of the gaps that remained are now closed, and closing them sharpened the conclusion rather than the guarantee. The nonlinear analogue is not a separate case: the two-action Jacobian determinant is $\text{gain}^2 dt^3$ identically for any $C^1$ force, so the same constant serves the cart and the pendulum, and it degenerates only on the mode clamp, where no density argument can work because the truth maps a neighborhood to a point. And the planner-versus-gate mismatch is measured rather than argued: the certified region carries 1.9% of the exploited planner's queries (7.8% for the larger one). What is left is therefore not a hole in the argument but its conclusion — a sound certificate on a region the planner barely visits. Certifying the planner's own visitation would need a gate that samples it, which is precisely the change the companion paper's closing prescription asks for.

**play_cost $> 1$ is a normalization artifact** (blind $<$ random), reported unclamped and explained; the headline claims never depend on the excess over 1.

## 11 Conclusion

The companion paper ended by diagnosing the gate failure as a reach-distribution shift and prescribing verification on the distribution the planner visits. This paper shows the diagnosis is not about discreteness. The gate-miss law, the identifiability argument, and the play-cost bound are measure-theoretic and survive the move to continuous state spaces intact; what the move changes is *where the localized failure can live* (hybrid mode boundaries — an omitted mode is a rare rule) and *who can express it* (programs, not smooth function classes: Proposition 8 and Table 11). On two minimal hybrid instruments the entire discrete phenomenology reproduces — threshold law, knob-invariant exploitation, the reach mechanism — with the gate's acceptance rate closely tracking $(1-r)^N$ at gate scale (within sampling noise; Section 5).

The synthesis experiment then delivers this paper's own finding: on the 1D instruments the continuous regime is the one where the danger law is *everything*. Given the mode in its sample, the LLM repairs it exactly — including from the synthesis examples alone — and when it fails to repair, the gate catches the superstitious patch; given the mode absent, every learner we tried (LLM, exact linear regression) is certified blind and the planner is exploited to below-random performance. That repair, though, is geometry-dependent: on a third, 4D bi-modal instrument with 2D modes it inverts — neither GPT-5.x size recovers the region rule from data (0/156, including a zero-curvature square and a guided 3×-budget treatment: a template prior over region forms, not curvature), and a second family fails in the same classes, and the danger law's exhaustiveness is scoped back to the 1D clamps, though the gate stays sound (Section 7.1). The provable core is the gate-miss and identifiability propositions (Section 3): the mode-absent event is a hole no learner can close *from the sample* — a prior or the specification could still supply the mode — and its probability is in closed form. Sample-covered exactness is the acceptance criterion itself; that accepted artifacts were also right off-sample — with the single phantom-mode exception, wrong only where its sample is silent — is an empirical regularity of the models tested (GPT-5.x repairs; Qwen stalls and is refused; Claude repairs most through a symmetry prior), not a theorem. For practitioners the message is one clause sharper than the discrete one: in continuous code-world-model pipelines, *sample coverage of the mode boundaries is the whole game* — the synthesis stack will translate what it is told and repair what it is shown, and no stage of it can know about what the sample never touched.

# A Reproducibility

All code is at https://github.com/JaviMaligno/code-world-models (`src/cwm/continuous/`, `scripts/continuous_*.py`); the experiment log is `docs/EXPERIMENTS.md` (sections prefixed "PAPER 2") and per-seed artifacts, including every synthesized program, are versioned under `results/`. CPU results (Tables 2–11):

```
PYTHONPATH=src python scripts/continuous_reach.py
                     # danger-curve table (tab:danger; ~2.5 min)
PYTHONPATH=src python scripts/continuous_axes.py
                     # axis-separation table (tab:axes; ~3 min)
PYTHONPATH=src python scripts/continuous_smooth_probe.py
                     # smooth-learner table (tab:smooth; ~11 s)
PYTHONPATH=src python scripts/continuous_pendulum.py
                     # pendulum table (tab:pendulum; ~2 min)
PYTHONPATH=src python scripts/continuous_cem.py
                     # second planner family (tab:cem; ~29 min)
PYTHONPATH=src python scripts/continuous_patch2d.py
                     # PatchField2D mechanism (tab:patch2d)
PYTHONPATH=src python scripts/continuous_cem_patch2d.py
                     # PatchField2D CEM rows (sec:cem; ~14 min)
PYTHONPATH=src python scripts/continuous_eps_sweep_patch2d.py
```

```
                     # PatchField2D eps rows (tab:eps-sweep text; ~8 min)
PYTHONPATH=src python scripts/continuous_mitigation_patch2d.py
                     # PatchField2D mitigation (tab:patch2d-mitigation)
python scripts/make_paper2_figures.py     # figures from the JSONs
python scripts/audit_paper2_numbers.py    # re-derive every table cell
                     # (and the counted prose claims) from the JSONs
```

LLM arms (Azure credentials in `.env`; the runbook — commands, cost, predictions, troubleshooting — is the section “Runbook — LLM arms” of the design spec under `docs/specs/`, dated 2026-07-06):

```
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5 --x-wall 4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 5
# Pendulum synthesis arm (second instrument; Azure credentials as above)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
  --instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument pendulum --th-stop 1.0
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
  --instrument pendulum --th-stop 1.0
# Qwen cross-family spot-checks
# (needs HF_TOKEN: HF Inference Providers router)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
  --compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
  --instrument pendulum --th-stop 1.4 \
  --compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
# Claude agent-relayed cross-family spot-checks (agent scaffold relays
# pipeline-identical messages; see scripts/continuous_claude_step.py,
# results/continuous_claude_relay.json + results/claude_relay_transcripts/).
# One relay round = init (writes msgN.txt) -> relay it -> check (gates the
# reply, writes msgN+1.txt or classifies). The 2D arm, per seed:
PYTHONPATH=src python scripts/continuous_claude_step.py init 10000 \
  results/claude_relay_transcripts --instrument patch2d --k1 3 --k2 7 \
  --arm incomplete
PYTHONPATH=src python scripts/continuous_claude_step.py check 10000 \
  results/claude_relay_transcripts 0 REPLYFILE --instrument patch2d \
  --k1 3 --k2 7 --arm incomplete --model-label "claude-sonnet-5 (agent-relayed)"
# Per-iteration ledger, re-derived from the versioned transcripts (CPU only):
PYTHONPATH=src python scripts/claude_relay_ledger.py \
  results/claude_relay_transcripts --tag patch2d_k3_7 --k1 3 --k2 7
# PatchField2D synthesis arm (4D bi-modal; two bi-knob cells)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument patch2d --k1 3 --k2 7
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
```

```
  --instrument patch2d --k1 3 --k2 7
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument patch2d --k1 5 --k2 9
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
  --instrument patch2d --k1 5 --k2 9
# Mitigation sweep (CPU only)
PYTHONPATH=src python scripts/continuous_mitigation.py
# eps-sensitivity sweep (CPU only)
PYTHONPATH=src python scripts/continuous_eps_sweep.py
```

## References


Javier Aguilar Martín. When a verified world model still loses: Play-adequacy vs prediction-accuracy in LLM-synthesized code world models, 2026. Companion paper (paper 1). arXiv:2607.14169, https://arxiv.org/abs/2607.14169; repository: https://github.com/JaviMaligno/code-world-models.

Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, pages 2669–2678, 2018.

Yashwanth Annpureddy, Che Liu, Georgios Fainekos, and Sriram Sankaranarayanan. S-TaLiRo: A tool for temporal logic falsification for hybrid systems. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, volume 6605 of *Lecture Notes in Computer Science*, pages 254–257. Springer, 2011.

Alberto Bemporad and Manfred Morari. Control of systems integrating logic, dynamics, and constraints. *Automatica*, 35(3):407–427, 1999.

Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Koen Claessen and John Hughes. QuickCheck: A lightweight tool for random testing of Haskell programs. In *Proceedings of the Fifth ACM SIGPLAN International Conference on Functional Programming (ICFP)*, pages 268–279, 2000.

Anthony Corso, Robert J. Moss, Mark Koren, Ritchie Lee, and Mykel J. Kochenderfer. A survey of algorithms for black-box safety validation of cyber-physical systems. *Journal of Artificial Intelligence Research*, 72:377–428, 2021.

Rémi Coulom. Efficient selectivity and backup operators in monte-carlo tree search. In *Computers and Games (CG 2006)*, volume 4630 of *Lecture Notes in Computer Science*, pages 72–83. Springer, 2006.

Nima Fazeli, Roman Kolbert, Russ Tedrake, and Alberto Rodriguez. Parameter and contact force estimation of planar rigid-bodies undergoing frictional contact. *The International Journal of Robotics Research*, 36(13–14):1437–1454, 2017.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799, 2023.

Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.

Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Levente Kocsis and Csaba Szepesvári. Bandit based Monte-Carlo planning. In *Proceedings of the 17th European Conference on Machine Learning (ECML)*, pages 282–293, 2006.

Nathan O. Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC)*, volume 120 of *Proceedings of Machine Learning Research*, pages 761–770, 2020. arXiv:2002.04523.

Wolfgang Lehrach, Daniel Hennes, Miguel Lazaro-Gredilla, Xinghua Lou, Carter Wendelken, Zun Li, Antoine Dedieu, Jordi Grau-Moya, Marc Lanctot, Atil Iscen, John Schultz, Marcus Chiam, Ian Gemp, Piotr Zielinski, Satinder Singh, and Kevin P. Murphy. Code World Models for general game playing. In *International Conference on Learning Representations (ICLR)*, 2026. OpenReview id 1UoB7IWiku.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 9493–9500, 2023.

Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, and Sergey Levine. Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

Simone Paoletti, Aleksandar Lj. Juloski, Giancarlo Ferrari-Trecate, and René Vidal. Identification of hybrid systems: A tutorial. *European Journal of Control*, 13(2-3):242–260, 2007.

James B. Rawlings, David Q. Mayne, and Moritz M. Diehl. *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2nd edition, 2017.

Reuven Y. Rubinstein and Dirk P. Kroese. *Simulation and the Monte Carlo Method*. Wiley, 3rd edition, 2017.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.